

Question #1 of 137

Question ID: 1472946

Free cash flow approaches are the *best* source of value when:

- A) return on assets is falling. 
- B) dividends are not paid. 
- C) a firm has significant minority interest. 

Explanation

Free cash flow approaches are best when dividends are not paid. Both remaining responses have nothing to do with the decision.

(Module 21.1, LOS 21.a)

Question #2 of 137

Question ID: 1473008

The stable-growth free cash flow to the firm (FCFF) model is *most* useful in valuing firms that:

- A) have capital expenditures that are significantly higher than depreciation. 
- B) are growing at a rate significantly lower than that of the overall economy. 
- C) have capital expenditures that are not significantly higher than depreciation. 

Explanation

The stable-growth FCFF model is useful for valuing firms that are expected to have growth rates close to that of the overall economy. Since the rate of growth approximates that for the overall economy, these firms should have capital expenditures that are not significantly different than depreciation.

(Module 21.5, LOS 21.j)

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Question ID: 1473009

Which of the following is *most* useful in analyzing firms that have high leverage and high growth?

- A) Two-stage free cash flow to the firm (FCFF) model. 
- B) Two-stage free cash flow to equity (FCFE) model. 
- C) Stable-growth free cash flow to the firm (FCFF) model. 

Explanation

Of the cash flow valuation models mentioned above, the two-stage FCFF model is most useful in analyzing the firms that have high leverage and high growth. The high growth will make the stable growth models inapplicable, while the high leverage makes the FCFF model more attractive.

(Module 21.5, LOS 21.j)

What is the *most likely* reason that you get an extremely low value from the three-stage FCFE model? Capital expenditures are significantly:

- A) higher than depreciation in the stable-growth phase. ✔
- B) less than depreciation during the high-growth phase. ✘
- C) higher than depreciation during the high-growth phase. ✘

Explanation

If capital expenditures estimates are significantly higher than depreciation for the stable growth period, then the three-stage FCFE model might result in an extremely low value. One possible solution for the problem is to grow the capital expenditures more slowly than depreciation in the transition period to narrow the difference. Another is to assume that capital expenditures and depreciation will offset when growth normalizes.

(Module 21.1, LOS 21.a)

Burcar-Eckhardt, a firm specializing in value investments, has been approached by the management of Overhaul Trucking, Inc., to explore the possibility of taking the firm private via a management buyout. Overhaul's stock has stumbled recently, in large part due to a sudden increase in oil prices. Management considers this an opportune time to take the company private. Burcar would be a minority investor in a group of friendly buyers.

Jaimie Carson, CFA, is a private equity portfolio manager with Burcar. He has been asked by Thelma Eckhardt, CFA, one of the firm's founding partners, to take a look at Overhaul and come up with a strategy for valuing the firm. After analyzing Overhaul's financial statements as of the most recent fiscal year-end (presented below), he determines that a valuation using Free Cash Flow to Equity (FCFE) is most appropriate. He also notes that there were no sales of PPE.

Overhaul Trucking, Inc.		
Income Statement		
April 30, 2005		
(Millions of dollars)		
	2005	2006E
Sales	300.0	320.0
Gross Profit	200.0	190.0
SG&A	50.0	50.0
Depreciation	<u>70.0</u>	<u>80.0</u>
EBIT	80.0	60.0
Interest Expense	30.0	34.0
Taxes (at 35 percent)	<u>17.5</u>	<u>9.1</u>
Net Income	32.5	16.9

Overhaul Trucking, Inc.	
Balance Sheet	
April 30, 2005	

(Millions of dollars)		
	2005	2006E
Cash	10.0	15.0
Accounts Receivable	50.0	55.0
Gross Property, Plant & Equip.	400.0	480.0
Accumulated Depreciation	(160.0)	(240.0)
Total Assets	300.0	310.0
Accounts Payable	50.0	70.0
Long-Term Debt	140.0	113.1
Common Stock	80.0	80.0
Retained Earnings	30.0	46.9
Total Liabilities & Equity	300.0	310.0

Eckhardt agrees with Carson's choice of valuation method, but her concern is Overhaul's debt ratio. Considerably higher than the industry average, Eckhardt worries that the firm's heavy leverage poses a risk to equity investors. Overhaul Trucking uses a weighted average cost of capital of 12% for capital budgeting, and Eckhardt wonders if that's realistic.

Question #5 - 8 of 137

Question ID: 1473065

Which of the following is one of the differences between FCFE and FCFF? FCFF does not deduct:

- A) operating expenses.
- B) working capital investment.
- C) interest payments to bondholders.



Explanation

FCFF includes the cash available to all of the firm's investors, including bondholders. Therefore, interest payments to bondholders are not removed from revenues to derive FCFF. FCFE is FCFF minus interest payments to bondholders plus net borrowings from bondholders.

(Module 21.5, LOS 21.i)

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Question ID: 1473066

Which of the following is the *least likely* reason for Carson's decision to use FCFE in valuing Overhaul rather than FCFF?

- A) Overhaul's capital structure is stable.
- B) FCFE is an easier and more straightforward calculation than FCFF.
- C) Overhaul's debt ratio is significantly higher than the industry average.



Explanation

The difference between FCFF and FCFE is related to capital structure and resulting interest expense. When the company's capital structure is relatively stable, FCFE is easier and more straightforward to use. FCFF is generally the best choice when FCFE is negative or the firm is highly leveraged. The fact that Overhaul's debt ratio is significantly higher than the industry average would argue against the use of FCFE. Hence, this is the least likely reason to favor FCFE.

(Module 21.5, LOS 21.i)

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Question ID: 1473067

What is the expected growth rate in FCFF that Carson must have used to generate his valuation of \$1.08 billion?

- A) 5%. 
- B) 7%. 
- C) 12%. 

Explanation

Since Firm Value = $FCFF_1 / (WACC - g)$, we first need to determine $FCFF_1$, which is FCFF in 2006: $FCFF = NI + NCC + [Int \times (1 - \text{tax rate})] - FCInv - WCInv$

$$= 16.9 + 80 + [34 \times (1 - 0.35)] - (480 - 400) - [(55 - 70) - (50 - 50)]$$

$$= 16.9 + 80 + 22.1 - 80 - (-15) = 54$$

$$\text{Firm Value} = FCFF_1 / (WACC - g)$$

$$1080 = 54 / (0.12 - x)$$

$$[(1080)(0.12)] - 1080x = 54$$

$$129.6 - 1080x = 54$$

$$75.6 = 1080x$$

$$0.07 = x$$

The expected growth rate in FCFF that Carson must have used is 7%.

(Module 21.5, LOS 21.i)

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Question ID: 1473068

If Carson had estimated FCFE under the assumption that Overhaul Trucking maintains a target debt-to-asset ratio of 36 percent for new investments in fixed and working capital, what would be his forecast of 2006 FCFE?

- A) \$26.5 million. 
- B) \$9.6 million. 
- C) \$16.9 million. 

Explanation

$$FCFE = NI - [(1 - DR) \times (FCInv - Dep)] - [(1 - DR) \times WCInv]$$

Where: DR = target debt to asset ratio

$$\begin{aligned} FCFE &= 16.9 - [(1 - 0.36) \times (480 - 400 - 80)] - [(1 - 0.36) \times ((55 - 70) - (50 - 50))] \\ &= 16.9 - (0.64 \times 0) - (0.64 \times (-15)) \\ &= 16.9 + 0 + 9.6 = 26.5 \end{aligned}$$

(Module 21.5, LOS 21.i)

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Question ID: 1472973

In forecasting free cash flows it is *most* common to assume that:

- A) historical levels of free cash flow will persist.
- B) the firm has no non-cash expenses.
- C) the firm capital structure is static.



Explanation

A firm's target debt ratio is usually assumed to remain constant. Historical cash flows are generally projected forward with a growth rate.

(Module 21.5, LOS 21.e)

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Question ID: 1472964

Free cash flow to the firm is equal to cash flow from operations minus fixed capital investment:

- A) minus after-tax interest expense.
- B) plus after-tax interest expense.
- C) minus pre-tax interest expense.



Explanation

Free cash flow to the firm is equal to cash flow from operations minus fixed capital investment plus after-tax interest expense.

(Module 21.2, LOS 21.c)

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Question ID: 1472986

Dividends paid out to the shareholders:

- A) may be higher than free cash flow to equity FCFE.
- B) are always less than free cash flow to equity (FCFE).
- C) are always equal to free cash flow to equity (FCFE).



Explanation

Dividends represent the cash that the firm chooses to pay to the shareholders and the amount of the dividend is subject to the discretion of the firm. Dividends can be equal to, lower or higher than FCFE. For example, sometimes firms may pay dividends in years when there is a net loss.

(Module 21.5, LOS 21.f)

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Question ID: 1473003

A biotech firm is currently experiencing high growth and pays no dividends. One of their product patents is scheduled to expire in 5 years. This firm would be a good candidate for which of the following valuation models?

- A) Two-stage dividend discount model (DDM).
- B) Two-stage free cash flow to equity (FCFE).
- C) Single-stage free cash flow to equity (FCFE).



Explanation

The two-stage FCFE model is well suited to value a firm that is currently experiencing high growth and will likely see this growth drop to a lower, more stable rate in the future.

(Module 21.5, LOS 21.j)

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Question ID: 1473006

Which of the following free cash flow to equity (FCFE) models is *most* suited to analyze firms in an industry with significant barriers to entry?

- A) Two-stage FCFE Model.
- B) FCFE Perpetuity Model.
- C) Stable Growth FCFE Model.



Explanation

The two-stage FCFE model is most suited for analyzing firms in high growth that will maintain that growth for a specific period, such as firms with patents or firms in an industry with significant barriers to entry.

(Module 21.5, LOS 21.j)

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Question ID: 1472951

Free cash flow (FCF) approaches are the *best* source of value when:

- A) a firm has no preferred stock.
- B) dividends are paid but do not reflect the company's capacity to pay dividends.
- C) a firm has significant minority interest.



Explanation

FCF approaches are best when dividends are paid but do not appear to be representative of the firm's capacity to pay them. Both remaining responses have nothing to do with the decision.

(Module 21.1, LOS 21.a)

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Question ID: 1472998

If the investment in fixed capital and working capital offset each other, free cash flow to the firm (FCFF) may be proxied by:

- A) earnings before interest and taxes (EBIT). 
- B) after-tax EBIT plus non-cash charges. 
- C) net income plus after-tax interest. 

Explanation

The answer is indicated by the definition of FCFF: $FCFF = EBIT(1 - \text{tax rate}) + \text{Dep} - \text{FCInv} - \text{WCInv}$, which assumes that depreciation is the only non-cash charge. Further: $FCFF = NI + \text{NCC} + \text{Int}(1 - \text{tax rate}) - \text{FCInv} - \text{WCInv}$.

(Module 21.5, LOS 21.h)

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Question ID: 1473033

A firm has projected free cash flow to equity next year of \$1.25 per share, \$1.55 in two years, and a terminal value of \$90.00 two years from now, as well. Given the firm's cost of equity of 12%, a weighted average cost of capital of 14%, and total outstanding debt of \$30.00 per share, what is the current value of equity?

- A) \$71.74. 
- B) \$74.10. 
- C) \$41.54. 

Explanation

Value of equity = $\$1.25 / (1.12)^1 + \$1.55 / (1.12)^2 + \$90.00 / (1.12)^2 = \74.10

(Module 21.5, LOS 21.k)

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Question ID: 1473012

Which of the following types of company is the E-Model, a three-stage free cash flow to equity (FCFE) Model, best suited for? Companies:

- A) with patents or firms in an industry with significant barriers to entry. 
- B) growing at a rate similar to or less than the nominal growth rate of the economy. 
- C) in high growth industries that will face increasing competitive pressures over time, leading to a gradual decline in growth to a stable level. 

Explanation

The three-stage FCFE model, or E-Model, is most suited to analyzing firms currently experiencing high growth that will face increasing competitive pressures over time, leading to a gradual decline in growth to a stable level. The two-stage model is best suited to analyzing firms in a high growth phase that will maintain that growth for a specific period, such as firms with patents or firms in an industry with significant barriers to entry. Companies growing at a rate similar to or less than the nominal growth rate of the economy are best suited for the Stable Growth FCFE Model. A firm that pays out all of its earnings as dividends will have a growth rate of zero (remember $g = RR \times ROE$) and would not be valued using the three-stage FCFE model.

(Module 21.5, LOS 21.j)

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Question ID: 1586163

In computing free cash flow, the most significant non-cash expense is usually:

A) deferred taxes.



B) capital expenditures.



C) depreciation.



Explanation

Depreciation is usually the largest non-cash expense.

(Module 21.1, LOS 21.c)

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Question ID: 1473032

Using the stable growth free cash flow to the firm (FCFF) model, what is the value of Quality Builders under the assumptions contained in the table below?

Quality Builders	
Free Cash Flow to the Firm	
Year 0	
EBIT	\$500
Depreciation	\$200
Capital Spending	\$300
Working Capital Additions	\$30
Tax Rate	40%
Assumed Constant Growth Rate in Free Cash Flow	5%
Weighted-average Cost of Capital	11%

A) \$6,475.00.



B) \$2,975.00.



C) \$2,833.33.



Explanation

The stable growth FCFF model assumes that FCFF grows at a constant rate forever. FCFF in Year 0 is equal to $\text{EBIT}(1 - \text{tax rate}) + \text{Depreciation} - \text{Capital Spending} - \text{Working Capital Additions} = 500(1 - 0.4) + 200 - 300 - 30 = 170$. The Firm Value = $\text{FCFF}_1 / (r - g_n) = 170(1.05) / (0.11 - 0.05) = \$2,975$.

(Module 21.5, LOS 21.k)

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Question ID: 1472969

SOX Inc. expects high growth in the next 4 years before slowing to a stable future growth of 3%. The firm is assumed to pay no dividends in the near future and has the following forecasted free cash flow to equity (FCFE) information on a per share basis in the high-growth period:

	Year 1	Year 2	Year 3	Year 4
FCFE	\$3.05	\$4.10	\$5.24	\$6.71

High-growth period assumptions:

- SOX Inc.'s target debt ratio is 40% and a beta of 1.3.
- The long-term Treasury Bond Rate is 4.0%, and the expected equity risk premium is 6%.

Stable-growth period assumptions:

- SOX Inc.'s target debt ratio is 40% and a beta of 1.0.
- The long-term Treasury Bond Rate is 4.0% and the expected equity risk premium is 6%.
- Capital expenditures are assumed to equal depreciation.
- In year 5, earnings are \$8.10 per share while the change in working capital is \$2.00 per share.
- Earnings and working capital are expected to grow by 3% a year in the future.

In year 5, what is the free cash flow to equity (FCFE) for SOX Inc.?

- A) \$7.30.
- B) \$6.10.
- C) \$6.90.



Explanation

In year 5, $\text{FCFE} = \text{Earnings per share} - (\text{Capital Expenditures} - \text{Depreciation})(1 - \text{Debt Ratio}) - (\text{Change in working capital})(1 - \text{Debt Ratio}) = 8.10 - 0(1 - 0.4) - 2.00(1 - 0.4) = 6.90$.

(Module 21.4, LOS 21.d)

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Question ID: 1472985

Which of the following statements is *least* accurate? A firm's free cash flows to equity (FCFE) is the cash available to stockholders after funding:

- A) capital expenditure requirements.



B) debt principal repayments.



C) dividend payments.



Explanation

A firm's FCFE is the cash available to stockholders after funding capital expenditures and debt principal repayments.

(Module 21.5, LOS 21.f)

TOY, Inc. is a company that manufactures dolls, games, and other items to entertain children.

The following table provides background information for TOY, Inc. on a per share basis in the year 0:

Current Information	Year 0
Earnings	\$5.00
Capital Expenditures	\$2.40
Depreciation	\$1.80
Change in Working Capital	\$1.70
Cost of equity	12.0%
Target debt ratio	30.0%
Market value of stock	\$56.00
Shares outstanding	5.0 million
Interest expense	\$7.2 million
Cash & short-term investments	\$40.0 million
Tax rate	37.5%

Earnings, capital expenditures, depreciation, and working capital are all expected to grow by 5.0% per year in the future.

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Question ID: 1473077

In year 1, the forecasted free cash flow to equity (FCFE) for TOY, Inc. is *closest* to:

A) \$3.56.



B) \$4.31.



C) \$4.53.



Explanation

FCFE year 0 = Earnings per share – [(Capital Expenditures – Depreciation) × (1 – Debt Ratio)] – [(Change in working capital) × (1 – Debt Ratio)] = 5.00 – [(2.40 – 1.80) × (1 – 0.30)] – [(1.70) × (1 – 0.30)] = 3.39.

FCFE for year 1 = FCFE year 0 × (1 + growth rate) = 3.39 × (1.05) = \$3.56.

(Module 21.5, LOS 21.m)

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Question ID: 1473078

The value of TOY, Inc.'s stock given the above assumptions, is *closest* to:

- A) \$50.86.
- B) \$64.71.
- C) \$61.57.

**Explanation**

The value of the stock = $FCFE_1 / (r - g_n) = 3.56 / (0.12 - 0.05) = 50.86$.

(Module 21.5, LOS 21.m)

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Question ID: 1473079

Comparing the current market value of TOY to our estimate of the stock's current market value, it is *most likely* that at the current market price of \$56.00, TOY Inc. stock is:

- A) overvalued.
- B) undervalued.
- C) fairly valued.

**Explanation**

Our calculated value of the stock = $FCFE_1 / (r - g_n) = 3.56 / (0.12 - 0.05) = \50.86 . The current market price is \$56.00, because the market price is greater than the estimated price, the stock is overvalued in the market.

(Module 21.5, LOS 21.m)

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Question ID: 1473080

Senior management of TOY Inc. is considering selling the company to a rival firm that has offered \$450 million. If the current market price represents the fair value of equity and TOY Inc. maintains its target capital structure, the bid represents a price that is:

- A) less than the total value of the firm.
- B) about the same total value of the firm.
- C) greater than the total value of the firm.

**Explanation**

The total value of a firm is the total market value of equity plus the total market value of debt. The total value of equity is \$56.00 per share $\times$ 5,000,000 shares = \$280 million. Equity represents 70.0% of the capital structure. The total value of the firm is thus \$280 million / 0.70 = \$400 million. An offer of \$450 million is a premium of \$50 million – a price greater than the current value of the firm.

(Module 21.5, LOS 21.m)

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Question ID: 1472963

Free cash flow to the firm (FCFF) adjusts earnings before interest and taxes (EBIT) by:

- A) deducting taxes, adding back depreciation, and deducting the investments in fixed capital and working capital. 
- B) subtracting investments in fixed capital and working capital. 
- C) adding taxes, deducting depreciation, and adding back the investments in fixed capital and working capital. 

Explanation

As presented in the reading: $FCFF = EBIT (1 - \text{tax rate}) + \text{Dep} - \text{FCInv} - \text{WCInv}$.

(Module 21.2, LOS 21.c)

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Question ID: 1472960

The ownership perspective implicit in the free cash flow to equity valuation approach is of:

- A) control. 
- B) a preferred stockholder. 
- C) a minority position. 

Explanation

Dividend policy can be changed by the buyer of a firm. Thus, the free cash flow perspective looks to the source of dividends in a position of control rather than directly at dividends.

(Module 21.1, LOS 21.b)

Question #28 of 137

Question ID: 1508664

The two-stage (stable growth) free cash flow to equity (FCFE) and free cash flow to the firm (FCFF) models typically assume:

- A) a high level of free cash flow for n years and then a lower level of free cash flow thereafter. 
- B) high growth in free cash flow for n years and then constant growth in free cash flow forever after. 
- C) growth of free cash flow that declines to the required rate of return in the last stage. 

Explanation

The two-stage model using either FCFE or FCFF typically assumes a high growth of free cash flow for n years and then a constant growth in free cash flow forever after. Multi-stage models assume that the required rate of return exceeds the growth rate in the last stage. In a two-stage free cash flow models, the growth rate in the second stage represents the long-run sustainable growth rate, which is generally a low rate that is close to the GDP growth rate.

(Module 21.5, LOS 21.j)

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Question ID: 1473005

The stable-growth free cash flow to equity (FCFE) model is best suited for which of the following types of companies? Companies:

- A) with patents that will not expire for 20 or more years. 
- B) growing at a rate similar or less than the nominal growth rate of the economy. 
- C) with significant barriers to entry. 

Explanation

Companies growing at a rate similar to or less than the nominal growth rate of the economy are best suited for the Stable Growth FCFE Model. The three-stage FCFE model is most suited to analyzing firms currently experiencing high growth that will face increasing competitive pressures over time, leading to a gradual decline in growth to a stable level. The two-stage model is best suited to analyzing firms in a high growth phase that will maintain that growth for a specific period, such as firms with patents or firms in an industry with significant barriers to entry.

(Module 21.5, LOS 21.j)

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Question ID: 1473011

The one-stage (stable growth) free cash flow models assume:

- A) the required rate of return exceeds the growth rate. 
- B) the required rate of return is less than the growth rate. 
- C) a constant growth rate for n years and a high growth rate forever thereafter. 

Explanation

The one-stage model using either free cash flow to equity (FCFE) or free cash flow to the firm (FCFF) assumes that the required rate of return exceeds the growth rate. If this was not the case, the model would produce an unrealistic negative price.

(Module 21.5, LOS 21.j)

Question #31 of 137

Question ID: 1473071

A firm has:

- Free cash flow to equity = \$4.0 million.
- Cost of equity = 12%.
- Long-term expected growth rate = 5%.
- Value of equity per share = \$57.14 per share.

What will happen to the value of the firm if free cash flow to equity decreases to \$3.2 million?

A) The value will increase.



B) There is insufficient information to tell.



C) The value will decrease.



Explanation

Everything else being constant, a decrease in free cash flow to equity should decrease the value of the firm.

(Module 21.5, LOS 21.i)

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Question ID: 1472940

In the stable-growth FCFE model, an extremely low value can result from all of the following *EXCEPT*:

A) the required rate of return is too high for a stable firm.



B) the expected growth rate is too high for a stable firm.



C) capital expenditures are too high relative to depreciation.



Explanation

If the expected growth rate is too high for a stable firm, the value obtained using the stable-growth FCFE model will be extremely high.

(Module 21.1, LOS 21.a)

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Question ID: 1472968

Using the information below, value the stock of Symphony Publishing, Inc. using the free cash flow from equity (FCFE) valuation method.

- Required return of 13.0%.
- Value at the end of year 3 of 13 times $FCFE_3$.
- Shares outstanding: 10.0 million.
- Net income in year 1 of \$10.0 million, projected to grow at 10% for the next two years.
- Depreciation per year of \$3.0 million.
- Capital Expenditures per year of \$2.5 million.
- Increase in working capital per year of \$1.0 million.
- Principal repayments on debt per year of \$1.5 million.

The value per share of Symphony Publishing is approximately:

A) \$112.10.



B) \$14.10.



C) \$11.21.



Explanation

Step 1: Calculate each year's FCFE and discount at the required return.

- FCFE = net income + depreciation – capital expenditures – increase in working capital – principal repayments + new debt issues
- Year 1: $10.0 + 3.0 - 2.5 - 1.0 - 1.5 = 8.0$,
- $PV = 7.08 = 8.0 / (1.13)^1$, or $FV = -8.0$, $I = 13$, $PMT = 0$, $N = 1$, Compute PV
- Year 2: $10.0 \times 1.10 + 3.0 - 2.5 - 1.0 - 1.5 = 9.0$,
- $PV = 7.05 = 9.0 / (1.13)^2$, or $FV = -9.0$, $I = 13$, $PMT = 0$, $N = 2$, Compute PV
- Year 3: $10.0 \times (1.10)^2 + 3.0 - 2.5 - 1.0 - 1.5 = 10.10$
- $PV = 7.00 = 10.10 / (1.13)^3$, or $FV = -10.10$, $I = 13$, $PMT = 0$, $N = 3$, Compute PV

Step 2: Calculate Present Value of final cash flow times FCFE multiple.

- Value at end of year 3 = $FCFE_3 \times \text{multiple} = 10.10 \times 13 = 131.30$
- $PV = 91.00 = 131.30 / (1.13)^3$, or using calculator, $N = 3$, $FV = -131.30$, $I = 13$, $PMT = 0$, Compute PV

Step 3: Calculate per share value.

- Add up PV of FCFE and end value and divide by number of shares outstanding
- $= (7.08 + 7.05 + 7.00 + 91.0) / 10.0 = 11.21$

(Module 21.4, LOS 21.d)

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Question ID: 1472983

The primary difference between the three-stage DDM and the FCFE model is:

A) growth rate assumptions.



B) the definition of cash flows.



C) cost of equity.



Explanation

The primary difference between the dividend discount models and the free cash flow from equity models lies in the definition of cash flows. The FCFE model uses residual cash flows after meeting all financial obligations and investment needs. The DDM uses a strict definition of cash flows to equity, that is, the expected dividends on the stock.

(Module 21.5, LOS 21.g)

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Question ID: 1472970

BOX Inc. earned \$4.55 per share last year. The firm had capital expenditures of \$1.75 per share and depreciation expense of \$1.05. BOX Inc. has a target debt ratio of 0.25.

	High-Growth Period	Transitional Period	Stable-Growth Period
Duration	2 Years	5 Years	
Earnings growth rate	45%	Will decline 8% per year to 5% in the stable-growth period	5%
Growth in Capital Expenditures	30%	Increases by 8% per year	Same as Depreciation
Growth in Depreciation	30%	Increases by 13% per year	Same as Capital Expenditures
Change in Working Capital	Given Below	Given Below	\$2.25 per share in Year 8
Shareholder Required Return	25%	15%	10%

	Yr 0	Yr 1	Yr 2	Yr 3	Yr 4	Yr 5	Yr 6	Yr 7
EPS	4.55	6.60	9.57	13.11	16.91	20.46	23.12	24.27
Capital Expenditures	1.75	2.28	2.96	3.19	3.45	3.73	4.02	4.35
Depreciation	1.05	1.37	1.77	2.01	2.27	2.56	2.89	3.27
Change in WC	0.90	1.10	1.40	1.60	1.80	2.00	2.20	2.10
FCFE			7.63	11.01	14.67	18.08	20.62	21.89

In year 1, what is the free cashflow to equity (FCFE) for BOX Inc.?

- A) \$5.09.
- B) \$3.35.
- C) \$6.10.



Explanation

Year 1 FCFE = Earnings per share – (Capital Expenditures – Depreciation) (1 – Debt Ratio) – Change in working capital (1 – Debt Ratio)

$$\text{Year 1 FCFE} = 6.60 - (2.28 - 1.37)(1 - 0.25) - (1.1)(1 - 0.25) = 5.09$$

(Module 21.4, LOS 21.d)

Question #36 of 137

Question ID: 1472961

Which of the following items is NOT subtracted from the net income to calculate free cash flow to equity (FCFE)?

- A) Interest payments to bondholders.



B) increase in accounts receivable.



C) Increase in fixed assets.



Explanation

Interest payments to bondholders are included in the income statement and are already subtracted to calculate net income.

(Module 21.1, LOS 21.c)

Question #37 of 137

Question ID: 1472990

Which of the following is *least likely* to change as the firm changes leverage?

A) Free cash flows to firm (FCFF).



B) Free cash flows to equity (FCFE).



C) Weighted average cost of capital (WACC).



Explanation

The FCFFs are normally unaffected by the changes in leverage, as these are the cash flows before the debt payments.

(Module 21.5, LOS 21.f)

Question #38 of 137

Question ID: 1473051

An analyst has prepared the following scenarios for Schneider, Inc.:

Scenario 1 Assumptions

- Tax Rate is 40%.
- Weighted average cost of capital (WACC) = 12%.
- Constant growth rate in free cash flow = 3%.
- Last year, free cash flow to the firm (FCFF) = \$30.
- Target debt ratio = 10%.

Scenario 2 Assumptions

- Tax Rate is 40%.
- Earnings before interest and taxes (EBIT), capital expenditures, and depreciation will grow at 15% for the next three years.
- After three years, the growth in EBIT will be 2%, and capital expenditure and depreciation will offset each other.
- Weighted average cost of capital (WACC) during high growth stage = 20%.
- Weighted average cost of capital (WACC) during stable growth stage = 12%.
- Target debt ratio = 10%.

Scenario 2 FCFF	Year 0 (last year)	Year 1	Year 2	Year 3	Year 4
EBIT	\$15.00	\$17.25	\$19.84	\$22.81	\$23.27
Capital Expenditures	6.00	6.90	7.94	9.13	
Depreciation	4.00	4.60	5.29	6.08	
Change in Working Capital	2.00	2.10	2.20	2.40	2.40
FCFF		5.95	7.06	8.25	11.56

Given the assumptions contained in Scenario 2, what is the value of the firm?

A) \$81.54.



B) \$70.39.



C) \$96.92.



Explanation

Use the two-stage FCFF model to value the firm. The Terminal Value of the firm as of Year 3 = $11.56 / (0.12 - 0.02) = 115.60$. The value = $5.95 / (1.20) + 7.06 / (1.20)^2 + (8.25 + 115.62) / (1.20)^3 = 81.54$.

(Module 21.5, LOS 21.k)

Question #39 of 137

Question ID: 1472995

Which of the following statements regarding dividends and free cash flow to equity (FCFE) is *least* accurate?

A) Required returns are higher in FCFE discount models than they are in dividend discount models, since FCFE is more difficult to estimate.



B) FCFE can be negative but dividends cannot.



- C) FCFE discount models usually result in higher equity values than do dividend discount models (DDMs).



Explanation

Although FCFE may be more difficult to estimate than dividends, the required return is based on the risk faced by the shareholders, which would be the same under both models.

(Module 21.5, LOS 21.f)

Question #40 of 137

Question ID: 1473004

The three-stage FCFE model might result in an extremely high value if:

- A) the growth rate in the stable-period is too low.
- B) the growth rate in the stable-period is too high.
- C) the growth rate in the stable-period is equal to that of GNP.



Explanation

If the growth rate in the stable-period is too high or the high-growth and transition periods are too long, the three-stage FCFE model might result in an extremely high value.

(Module 21.5, LOS 21.j)

An analyst has prepared the following scenarios for Schneider Inc.:

Scenario 1 Assumptions:

- Tax Rate is 40%.
- Weighted average cost of capital (WACC) = 12.0%.
- Constant growth rate in free cash flow (FCF) = 3.0%.
- Year 0, free cash flow to the firm (FCFF) = \$30.0 million
- Target debt ratio = 10.0%.

Scenario 2 Assumptions:

- Tax Rate is 40.0%.
- Earnings before interest and taxes (EBIT), capital expenditures, and depreciation will grow at 20.0% for the next three years.
- After three years, the growth in EBIT will be 2.0%, and capital expenditure and depreciation will offset each other.
- Weighted average cost of capital (WACC) = 12.0%
- Target debt ratio = 10.0%.

Scenario 2 FCFF (in \$ millions)	Year 0	Year 1	Year 2	Year 3	Year 4
EBIT	\$45.00	\$54.00	\$64.80	\$77.76	\$79.70
Capital Expenditures	18.00	21.60	25.92	31.10	
Depreciation	12.00	14.40	17.28	20.74	
Change in Working Capital	6.00	6.30	6.60	7.20	7.20
FCFF		18.90	23.64	29.09	40.62

Other financial items for Schneider Inc.:

Estimated market value of debt = \$35.0 million

Cost of debt = 5.0%

Shares outstanding = 20 million.

Question #41 - 44 of 137

Question ID: 1508666

Given the assumptions contained in Scenario 1, the value of the firm is *most* accurately estimated as:

A) \$343 million.



B) \$333 million.



C) \$250 million.



Explanation

Under the stable growth FCFF model, the value of the firm = $FCFF_1 / (WACC - g_n) = \$30 \text{ million} \times (1.03) / (0.12 - 0.03) = \343.33 million .

(Module 21.5, LOS 21.k)

Question #42 - 44 of 137

Question ID: 1473019

In Scenario 2, the value of the firm is *closest* to:

A) \$315 million.



B) \$346 million.



C) \$321 million.



Explanation

The value of the firm is the present value of Year 1-3 plus the terminal value. The terminal value is: $FCFF \text{ for year 4} / (WACC - \text{growth rate}) = \$40.62 / (0.12 - 0.02) = \406.22 million in terms of year 3 dollars. The calculator inputs to solve NPV for the value of the firm is: CF0 = \$0, CF1 = \$18.90, CF2 = \$23.64, CF3 = \$29.09 + \$406.22 = \$435.31, I = 12. NPV = \$345.57 million.

(Module 21.5, LOS 21.k)

Question #43 - 44 of 137

Question ID: 1473020

The cost of equity for Schneider Inc. is *closest* to:

A) 13.0%.



B) 5.8%.



C) 11.3%.



Explanation

The weighted average cost of capital formula is $WACC = w_d \times r_d \times (1 - t) + w_e \times r_e$. The weight of debt is 10.0% – the weight of equity must be 90.0%.

$$0.12 = 0.10 \times 0.05 \times (1 - 0.40) + 0.90 \times r_e$$

$$0.120 - 0.003 = 0.90 \times r_e$$

$$0.117/0.9 = r_e$$

$$r_e = 13.0\%$$

(Module 21.5, LOS 21.k)

Question #44 - 44 of 137

Question ID: 1508667

The market value of Schneider Inc.'s stock is:

A) \$17.50 per share.



B) \$31.50 per share.



C) \$15.75 per share.



Explanation

The estimated market value of debt is \$35 million, which represents 10.0% of the value of the firm. The other 90.0% is the value of equity or \$315 million. \$315 million/20 million shares = \$15.75 per share.

(Module 21.5, LOS 21.k)

Ashley Winters, CFA, has been hired to value Goliath Communications, a company that is currently experiencing rapid growth and expansion. Winters is an expert in the communications industry and has had extensive experience in valuing similar firms. She is convinced that a value for the equity of Goliath can be reliably obtained through the use of a three-stage free cash flow to equity (FCFE) model with declining growth in the second stage. Based on up-to-date financial statements, she has determined that the current FCFE per share is \$0.90. Winters has prepared a forecast of expected growth rates in FCFE as follows:

Stage 1:	10.5% for years 1 through 3
Stage 2:	8.5% in year 4, 6.5% in year 5, 5.0% in year 6
Stage 3:	3.0% in year 7 and thereafter

Moreover, she has determined that the company has a beta of 1.8. The current risk-free rate is 3.0%, and the equity risk premium is 5.0%.

Other financial information:

Outstanding shares	10 million
Tax rate	40.0%
Interest expense	\$750,000
Net borrowing	−\$100,000
Cost of debt	7.5%

Debt-to-equity ratio	25.0%
Estimated growth rate for the firm	4.0%

Question #45 - 48 of 137

Question ID: 1473047

The terminal value in year 6 is *closest* to:

- A) \$16.86.
- B) \$21.68.
- C) \$25.29.



Explanation

Estimates for the future FCFE based on supplied growth rates are:

Year	1	2	3	4	5	6	7
Growth rate	10.5%	10.5%	10.5%	8.5%	6.5%	5.0%	3.0%
FCFE/share	\$0.995	\$1.099	\$1.214	\$1.318	\$1.403	\$1.473	\$1.518

$$R_5 = 1.518 / (12.0\% - 3.0\%) = 16.861$$

(Module 21.5, LOS 21.k)

Question #46 - 48 of 137

Question ID: 1473048

The per-share value Winters should assign to Goliath's equity is *closest* to:

- A) \$20.24.
- B) \$13.55.
- C) \$16.87.



Explanation

We find the value of the equity/share by discounting all future FCFE/share by the required rate of return on equity.

$$\text{value of equity/share} = \frac{0.995}{(1.12)} + \frac{1.099}{(1.12)^2} + \frac{1.214}{(1.12)^3} + \frac{1.318}{(1.12)^4} + \frac{1.403}{(1.12)^5} + \frac{1.473 + 16.867}{(1.12)^6} = \$13.55/\text{share}$$

Using our calculator, enter CF₀ = 0; C01 = 0.995; C02 = 1.099; C03 = 1.214; C04 = 1.318; C05 = 1.403; C06 = 1.473 + 16.867 = 18.34; I = 12; Compute → NPV = 13.55.

(Module 21.5, LOS 21.k)

Question #47 - 48 of 137

Question ID: 1473049

The weighted average cost of capital (WACC) is *closest* to:

- A) 10.5%.



B) 10.9%.



C) 11.1%.



Explanation

The debt-to-equity ratio of 25.0% means that the debt-to-total value is 25.0%/125.0% or 20.0%. The weight of debt is thus 20.0% and the weight of equity is 80.0%.

$$\text{The WACC} = [0.20 \times (0.075) \times (1 - 0.40)] + (0.80 \times 0.12) = 10.5\%$$

(Module 21.5, LOS 21.k)

Question #48 - 48 of 137

Question ID: 1473050

The value of the firm, based on the constant growth model, is *closest* to:

A) \$153 million.



B) \$140 million.



C) \$124 million.



Explanation

The estimated FCFF for year 0 is \$9.55 million and the WACC is 10.5% as calculated. If the growth rate for the firm is estimated as 4.0%, the value of the firm is:

$$\$9.55 \text{ million} \times (1.04)/(0.105 - 0.04) = \$152,800,000.$$

(Module 21.5, LOS 21.k)

Question #49 of 137

Question ID: 1472959

The ownership perspective implicit in the dividend valuation approach is of:

A) a preferred stockholder.



B) control.



C) a common stockholder.



Explanation

Dividends are most relevant to the stockholders who receive them and who have little control over their amount.

(Module 21.1, LOS 21.b)

Question #50 of 137

Question ID: 1472972

On a per share basis for a firm:

- Sales are \$10.00.
- Earnings per share (EPS) is \$4.00.
- Depreciation is \$3.00.
- After-tax interest is \$2.40.
- Investment in working capital is \$1.50.
- Investment in fixed capital is \$2.00.

What is the firm's expected free cash flow to the firm (FCFF) per share?

A) \$5.90.



B) \$7.50.



C) \$2.90.



Explanation

$FCFF = EPS + \text{net non-cash charges} + \text{after-tax interest} - FCInv - WCInv$

$FCFF = \$4.00 + 3.00 + \$2.40 - \$2.00 - 1.50 = \5.90

(Module 21.4, LOS 21.d)

Question #51 of 137

Question ID: 1473015

The two-stage FCFE model is suitable for valuing firms that:

A) have very high but declining growth rate in the initial stage.



B) have moderate growth in the initial phase that declines gradually to a stable rate.



C) are in an industry with significant barriers to entry.



Explanation

The two-stage FCFE model is suitable for valuing firms in industries with significant barriers to entry. Where these are present it is possible for the firm to maintain a high growth rate during an initial phase of low competition, and that the rate will drop sharply to a normalized rate when competition ultimately appears.

(Module 21.5, LOS 21.j)

William Bolton is an avid disc golf player and the owner of Deep in the Game Discs (DITGD), a business involved in wholesale distribution of discs and other disc golf equipment. DITGD supplies smaller outlets within the U.S. market and exports overseas. Will has built his business organically over a 20-year period, starting as a hobby but developing into a mid-sized business.

Will has recently lost some UK export customers to a smaller UK located competitor called Fishy Discs Ltd. Will recently met Neil Prebble, the owner of Fishy Discs at a trade fair and was considering a friendly acquisition in order to expand his business into the UK market.

Will has employed an accountancy firm with a corporate finance division, to give him some indication of a price to offer for Fishy Discs.

Exhibit 1: Fishy Discs—Selected Financial Information

Income Statement Period Ended 31st December 20x9

	£	£
Sales revenue		200,000
Expenses:		
Cost of goods sold	80,000	
Salaries	10,000	
Depreciation	14,000	
Interest	<u>1,000</u>	
		<u>105,000</u>
		95,000
Gain from sale of PP&E		<u>20,000</u>
Pre-tax income		115,000
Provision for taxes		<u>40,000</u>
Net income		<u>75,000</u>

Balance Sheet Year Ended 20x9

	20x8	20x9
	£	£

Current Assets

Cash	18,000	66,000
Accounts receivable	18,000	20,000
Inventory	14,000	10,000

Non-Current Assets

Gross PP&E	282,000	312,000
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Accumulated depreciation	(80,000)	(84,000)
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Total Assets	<u>252,000</u>	<u>324,000</u>
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Current Liabilities

Accounts payable	10,000	18,000
------------------	--------	--------

Salaries payable	16,000	9,000
------------------	--------	-------

Interest payable	6,000	7,000
------------------	-------	-------

Taxes payable	8,000	10,000
---------------	-------	--------

Dividends payable	2,000	12,000
-------------------	-------	--------

Non-Current Liabilities

Long term debt	20,000	30,000
----------------	--------	--------

Deferred tax	30,000	40,000
--------------	--------	--------

Stockholders' Equity

Contributed capital	100,000	80,000
---------------------	---------	--------

Retained earnings	60,000	118,000
-------------------	--------	---------

Liabilities and Equity	<u>252,000</u>	<u>324,000</u>
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On review of the PP&E footnote disclosure, it is discovered that equipment with a carrying value of £10,000 had been disposed of during 20x9.

All long-term debt is issued with a coupon such that the debt will trade at par value on issuance.

The effective tax rate relating to Fishy Discs is 40%.

Deferred tax liabilities are not expected to reverse for the foreseeable future.

The corporate finance firm employed by Will has decided to value Fishy Discs based on sustainable free cash flow, after removing one off items from the cash flow statement. In addition, they have considered how long Fishy Discs will be able to maintain its cash flow growth rates. Fishy Discs currently is the only domestically located UK supplier of disc golf equipment. Their results are included in Exhibit 2.

Exhibit 2: Fishy Discs Ltd. Valuation Data

Free cash flow to the firm	£75,000
(base period estimate)	
High growth rate	12%
Sustainable growth rate	4%
Duration of high growth	4 years
Declining growth duration	6 years
Cost of equity	10%
WACC	8%

The corporate finance team believes the market value of Fishy Discs debt is close to book value in the 20x9 accounts. The team believes that the decline in growth from 12% down to 4% will be linear.

Tony Cermak is a young modeler in the corporate finance team and he has raised a couple of comments regarding the valuation figures prepared for Fishy Discs.

Concern 1: Fishy Discs reduced its inventory between 20x8 and 20x9. This led to a boost of £4,000 in cash flows in 20x9. Given inventory, levels cannot decline below zero and we are forecasting Fishy Discs to grow, any boost to cash flow from inventory reduction is likely to be transitory and should be removed from sustainable cash flow.

Concern 2: Fishy Discs current high growth rates are linked to an exclusivity agreement that Prebble has with a U.S. disc producer. This agreement gives Fishy Discs sole supplier status for the global number one selling brand in the UK for a four-year period. At the end of this period, the U.S. supplier has indicated that other firms will be allowed to import and retail these products in the UK market. Given this, I believe our growth rate assumptions detailed in Exhibit 2 are unrealistic.

Tony has been given the firm's free cash flow valuation model guide to study before he attempts to value Fishy Discs. In particular, he is interested in the following formulas that have been given:

Exhibit 3: Cash Flow Valuation Guide Extract

FCFF from EBIT:

$$\text{FCFF} = \text{EBIT}(1 - T) + \text{depreciation} - \text{FC}_{\text{INV}} - \text{WC}_{\text{INV}}$$

FCFF from EBITDA:

$$\text{FCFF} = \text{EBITDA}(1 - T) + \text{depreciation} - \text{FC}_{\text{INV}} - \text{WC}_{\text{INV}}$$

Using the information available in Exhibit 1, Operating Cash Flow (CFO) for Fishy Discs is *closest* to?

A) £73,000.



B) £75,000.



C) £85,000.



Explanation

$$\text{CFO} = \text{NI} + \text{NCC} - \text{WC}_{\text{INV}}$$

Non-Cash Charges

+ Depreciation 14,000

– Gain on asset disposal (20,000)

+ Δ Deferred tax liability 10,000

Net impact 4,000

Working Capital Investment

	20x9	20x8
CA – cash and investments	30,000	32,000
CL – debt instruments and dividends payable	44,000	40,000
Working capital	<u>(14,000)</u>	<u>(8,000)</u>
Δ in working capital	–£6,000	

Note that the change in the deferred tax liability (DTL) is only included, as it is not expected to reverse. A DTL that is expected to reverse in the near term would be ignored. Whilst the DTL represents a boost to operating cash flows when it is created, it will reduce cash flows when it reverses. These two cash flow effects off set and as a result, it is best to ignore the DTL when estimating free cash flow if it is expected to reverse in the short run.

$$\text{CFO} = 75,000 + 4,000 + 6,000 = £85,000$$

(Module 21.4, LOS 21.d)

Question #53 - 57 of 137

Question ID: 1473059

Using the information available in Exhibit 1, capital expenditure for Fishy Discs is *closest* to?

A) £20,000.



B) £30,000.



C) £50,000.



Explanation

$FC_{INV} = \text{change in carrying value} + \text{dep}^n \text{ expense} - \text{gain on disposal}$

$$FC_{INV} = 26,000 + 14,000 - 20,000 = 20,000$$

Alternatively

Compute additions to PP&E as a residual figure:

FC_{INV} computation

PP&E 20x8	202,000	Proceeds (plug)	30,000
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Depreciation	(14,000)	Carrying value	<u>(10,000)</u>
--------------	----------	----------------	-----------------

Disposal	(10,000)	Gain/(loss)	20,000
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Additions (plug)	<u>50,000</u>
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PP&E 20x9	228,000
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$$FC_{INV} = £50,000 - £30,000 = £20,000$$

Note that additions and proceeds have been computed as residual (balancing) figures.

(Module 21.4, LOS 21.d)

Question #54 - 57 of 137

Question ID: 1473060

Assuming a CFO figure of £75,000 and capital expenditure of £20,000, Fishy Discs free cash flow to the firm for 20x9 is *closest* to?

A) £55,600.



B) £65,000.



C) £75,600.



Explanation

$$FCFF = \text{CFO} + \text{interest} (1 - T) - FC_{INV}$$

$$FCFF = £75,000 + £1,000 (1 - 0.4) - £20,000 = £55,600$$

(Module 21.4, LOS 21.d)

Question #55 - 57 of 137

Question ID: 1473061

Using only the corporate finance firm's data in Exhibit 2 and their growth assumptions, the value of Fishy Discs Ltd.'s equity is *closest* to?

A) £2,033,000.



B) £3,075,000.



C) £3,105,000.



Explanation

Using the 3 stage DCF model:

$$CF1 = £84,000$$

$$CF2 = £94,080$$

$$CF3 = £105,370$$

$$CF4 = £118,014 + £708,084 + £3,068,364$$

$$I = 8 \text{ CPT} \rightarrow NPV = £3,104,628$$

$$\text{Firm value} = £3,104,628$$

$$\text{Equity value} = \text{firm value} - \text{debt value}$$

$$£3,074,628 = £3,104,628 - £30,000$$

(Module 21.5, LOS 21.k)

Question #56 - 57 of 137

Question ID: 1473062

How many on of Tony's concerns are valid?

A) Both.



B) Neither.



C) Only concern 2.



Explanation

Concern 1: He is correct that declines in inventory give one off boosts to cash flow from operations and hence free cash flow. In addition, it is unlikely that Fishy Discs inventory will decline in future periods given we are expecting growth of 12% in the near term.

Concern 2: The corporate finance firm are using a three-stage model with declining growth after four years. The model takes six years for growth to decline to sustainable levels. Given that Fishy Discs will lose its exclusivity agreement with the U.S. producer in four years, it is likely that the decline in growth will be far more rapid. Once barriers to entry are removed growth will decline from 12% to 4% far more rapidly than is being modelled. For companies that sustain economic profit due barriers to entry such as patents, copyrights and other agreements using a two stage model may model the decline in growth due to an influx in competition more accurately.

(Module 21.5, LOS 21.j)

Question #57 - 57 of 137

Question ID: 1473063

How many of the FCFF definitions, in Exhibit 3, that Tony is studying are accurate?

A) Both.



B) Neither.



C) Only FCFF from EBIT.



Explanation

$$\text{FCFF} = \text{EBITDA}(1 - T) + (D \times T) - \text{FC}_{\text{INV}} - \text{WC}_{\text{INV}}$$

Note that the formulae given in the question is incorrect as it adds back depreciation rather than the tax shield on depreciation. It should be noted that other NCC might need to be adjusted for in practice.

(Module 21.5, LOS 21.h)

Question #58 of 137

Question ID: 1472984

The estimate of value from FCFE models will always be different than the value obtained using DDM, if:

A) FCFE is higher than dividends.



B) FCFE is greater than dividends, and the excess is not invested in zero NPV projects.



C) FCFE is higher than dividends, and the excess is invested in zero NPV projects.



Explanation

The estimate of value from FCFE models will always be different from the value obtained using DDM, if the FCFE is greater than dividends, and the excess cash is not invested in zero NPV projects.

(Module 21.5, LOS 21.g)

Question #59 of 137

Question ID: 1473075

Terminal value in a multi-stage free cash flow to equity (FCFE) valuation model is often calculated as the present value of:

A) a two-stage valuation model's price.



B) free cash flow divided by the growth rate.



C) FCFE divided by the total of required rate on equity minus growth.



Explanation

Terminal values are usually calculated as the present value of the price produced by a constant-growth model as of the beginning of the last stage, which is $\text{FCFE} / (\text{required rate on equity} - \text{growth})$.

(Module 21.5, LOS 21.l)

Question #60 of 137

Question ID: 1473007

Which of the following statements regarding the FCFF models is *most* accurate? The two-stage FCFF model is more useful than the stable-growth FCFF model when the firm is growing at a rate:

- A) significantly lower than that of the overall economy. 
- B) not significantly higher than that of the overall economy. 
- C) significantly higher than that of the overall economy. 

Explanation

The two-stage FCFF model is more useful in valuing a firm that is growing at a rate significantly higher than the overall economy. Since this cannot persist indefinitely, growth will eventually slow to a stable growth rate consistent with that of the economy.

(Module 21.5, LOS 21.j)

Question #61 of 137

Question ID: 1472994

Ignoring any costs related to financial distress, if a firm increases its financial leverage, the value of the firm should:

- A) increase because the weighted average cost of capital will be lower due to interest tax shields. 
- B) decrease because the required rate of return on debt is lower than that of equity. 
- C) increase because the FCFF will increase. 

Explanation

When a firm adds leverage, its value may increase due to the tax shields on interest expense and the generally lower cost of debt. In theory, there is an optimal capital structure. If the amount of debt employed is greater than the optimal, the costs associated with risk of bankruptcy or financial distress begin to outweigh the advantage of interest tax shields.

(Module 21.5, LOS 21.f)

Question #62 of 137

Question ID: 1473014

In using FCFE models, the assumption of growth should be:

- A) only consistent with the assumptions of capital spending and depreciation. 
- B) independent from the assumptions of other variables. 
- C) consistent with assumptions of other variables. 

Explanation

The assumption of growth should be consistent with assumptions about other variables. Net capital expenditures (capital expenditures minus depreciation) and beta (risk) used to calculate required rate of return should be consistent with assumed growth rate.

(Module 21.5, LOS 21.j)

An analyst has prepared the following scenarios for Schneider, Inc.:

Scenario 1 Assumptions:

- Tax rate is 40%.
- Weighted average cost of capital (WACC) = 12%.
- Constant growth rate in free cash flow = 3%.
- Last year, free cash flow to the firm (FCFF) = \$30.
- Target debt ratio = 10%.

Scenario 2 Assumptions:

- Tax rate is 40%.
- Expenses before interest and taxes (EBIT), capital expenditures, and depreciation will grow at 15% for the next three years.
- After three years, the growth in EBIT will be 2%, and capital expenditure and depreciation will offset each other.
- WACC during high growth stage = 20%.
- WACC during stable growth stage = 12%.
- Target debt ratio = 10%.

Scenario 2 FCFF	Year 0 (last year)	Year 1	Year 2	Year 3	Year 4
EBIT	\$15.00	\$17.25	\$19.84	\$22.81	\$23.27
Capital Expenditures	6.00	6.90	7.94	9.13	
Depreciation	4.00	4.60	5.29	6.08	
Change in Working Capital	2.00	2.10	2.20	2.40	2.40
FCFF		5.95	7.06	8.25	11.56

Assuming that Schneider, Inc., slightly increases its financial leverage, what should happen to its firm value? The firm value should:

- A)** increase due to the additional value of interest tax shields. 
- B)** decline due to the increase in risk. 
- C)** not change because financial leverage has no relationship with firm value. 

Explanation

For small changes in leverage, the additional value added by the interest tax shields will more than offset the additional risk of bankruptcy / financial distress. Given the tax advantage of debt, the firm's WACC should decline, not increase with small changes in leverage.

(Module 21.5, LOS 21.f)

Assuming that the investment in fixed capital and working capital offset each other, free cash flow to the firm (FCFF) may be proxied by net income if:

- A) non-cash charges and interest charges are equal. 
- B) non-cash charges and interest charges are zero. 
- C) earnings before interest and taxes (EBIT) equals depreciation. 

Explanation

The answer is shown by the relationship between FCFF and net income: $FCFF = NI + NCC + \text{Int} (1 - \text{tax rate}) - \text{FCInv} - \text{WCInv}$. Further: $FCFF = \text{EBIT} (1 - \text{tax rate}) + \text{Dep} - \text{FCInv} - \text{WCInv}$, which assumes that depreciation is the only non-cash charge.

(Module 21.5, LOS 21.h)

Question #65 of 137

Question ID: 1472942

Mark Washington, CFA, uses a two-stage free cash flow to equity (FCFE) discount model to value Texas Van Lines. His analysis yields an extremely low value, which he believes is incorrect. Which of the following is *least likely* to be a cause of this suspect valuation estimate?

- A) The forecast of working capital as a percentage of revenues in the stable growth period is not large enough to maintain the long-term sustainable growth rate. 
- B) Earnings are temporarily depressed because of a one-time extraordinary accounting charge in the most recent fiscal year. 
- C) The cost of equity estimate in the stable growth period is too high for a stable firm. 

Explanation

The larger the estimate of working capital as a percentage of revenues, the larger the investment in net working capital, and the lower the FCFE in the stable period. A low stable-period FCFE estimate will result in a low estimate of value today. The solution is to use a working capital ratio closer to the long-run industry average.

If the cost of equity estimate in the stable growth period is too high, the terminal value will be too low. Because the terminal value typically makes up a large portion of the current value, this will cause the current value estimate to be too low. The solution is to use a cost of equity estimate based on a beta of one.

If earnings are temporarily depressed, all the FCFE estimates will be low, and the current value estimate will be low. The solution is to use an estimate of long-run normalized earnings.

(Module 21.1, LOS 21.a)

Question #66 of 137

Question ID: 1473002

Which of the following statements about the three-stage FCFE model is *most* accurate?

- A) There is a transition period where the growth rate declines. 
- B) There is a final phase when growth rate starts to decline. 
- C) There is a transition period where the growth rate is stable. 

Explanation

In the three-stage FCFE model, there is an initial phase of high growth, a transition period where the growth rate declines, and a steady-state period where growth is stable.

(Module 21.5, LOS 21.j)

Question #67 of 137

Question ID: 1472974

In forecasting free cash flows it is common to assume that investment in working capital:

A) is greater than fixed capital investment during a growth phase.



B) will be financed using the target debt ratio.



C) will equal fixed capital investment.



Explanation

It is usually assumed that the investment in working capital will be financed consistent with the target debt ratio.

(Module 21.5, LOS 21.e)

The following information was collected from the financial statements of Hiller GmbH, a German consulting company, for the year ending December 31, 2013:

- Earnings per share = €4.50.
- Capital Expenditures per share = €3.00.
- Depreciation per share = €2.75.
- Increase in working capital per share = €0.75.
- Debt financing ratio = 30.0%.
- Cost of equity = 12.0%.
- Cost of debt = 6.0%.
- Tax rate = 30.0%.
- Outstanding shares = 100 million.
- New debt borrowing = €15.0 million.
- Debt repayment = €30.0 million.
- Interest expense = €7.1 million.

The financial leverage for the firm is expected to be stable. Hiller uses IFRS accounting standards and records interest expense as cash flow from financing (CFF).

Two analysts are valuing Hiller stock; both are basing their analysis on FCFE approaches.

Analyst #1 remarks: "Hiller is a relatively mature company; a constant growth model is the better approach."

Analyst #1 estimates FCFE based on the information above and a growth rate of 5.0%.

Analyst #2 states: "Hiller just acquired a rival that should change their growth pattern. I think a three stage growth model based on industry growth patterns should be used."

Analyst #2 estimates FCFE per share as €3.85. Growth rate estimates are listed below, and from year 7 and thereafter the estimated growth rate is 3.0%.

	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7+
Growth rates	12.5%	12.5%	12.5%	8.0%	6.5%	5.0%	3.0%

Question #68 - 71 of 137

Question ID: 1472978

Assuming a constant debt-to-asset ratio, the base year FCFE is *closest* to:

- A) €3.00.
- B) €3.80.
- C) €4.85.

**Explanation**

Base-year FCFE = EPS – (capital expenditures – depreciation) × (1 – debt ratio) – increase in working capital × (1 – debt ratio) = €4.50 – (€3.00 – €2.75)(1 – 0.30) – €0.75(1 – 0.30) = €3.80.

(Module 21.5, LOS 21.e)

Question #69 - 71 of 137

Question ID: 1472979

Using the stable-growth FCFE model as suggested by Analyst #1, the value of Hiller stock is *closest* to:

- A) €51.58.
- B) €54.29.
- C) €57.00.

**Explanation**

Value per share = (€3.80 × 1.05) / (0.12 – 0.05) = €57.00.

(Module 21.5, LOS 21.e)

Question #70 - 71 of 137

Question ID: 1472980

Based on Analyst #2's estimates, the sum of the terminal value plus the FCFE for year 6 is *closest* to:

- A) €75.80.
- B) €60.70.
- C) €82.40.

**Explanation**

Estimates for the future FCFE based on supplied growth rates are:

Year	0	1	2	3	4	5	6	7
Growth rate		12.5%	12.5%	12.5%	8.0%	6.5%	5.0%	3.0%
FCFE/share	€3.850	€4.331	€4.873	€5.482	€5.893	€6.335	€6.620	€6.818

Terminal value year 6 = $6.818 / (12.0\% - 3.0\%) = €75.76$

The nominal cash flow for year 6 is $€75.76 + €6.62 = €82.38$, which is the terminal cash flow plus the FCFE value for the year.

(Module 21.5, LOS 21.e)

Question #71 - 71 of 137

Question ID: 1472981

Based on Analyst #2's estimates, the value of Hiller stock is *closest* to:

A) €60.70.



B) €59.70.



C) €57.00.



Explanation

Year	0	1	2	3	4	5	6	7
Growth rate		12.5%	12.5%	12.5%	8.0%	6.5%	5.0%	3.0%
FCFE/share	€3.850	€4.331	€4.873	€5.482	€5.893	€6.335	€6.620	€6.818

Terminal value year 6 = $6.818 / (12.0\% - 3.0\%) = €75.76$

For the calculator find NPV: CF0 = 0, CF1 = €4.33, CF2 = €4.87, CF3 = €5.48, CF4 = €5.89, CF5 = €6.34, CF6 = €82.38, I/Y = 12. The result is €60.73.

(Module 21.5, LOS 21.e)

Question #72 of 137

Question ID: 1472949

Free cash flow to equity valuation uses which discount rate?

A) Cost of equity.



B) After-tax cost of debt.



C) Weighted average cost of capital.



Explanation

Free cash flow to equity valuation uses the opportunity cost relevant to stockholders, which is the cost of equity.

(Module 21.1, LOS 21.a)

Beachwood Builders merged with Country Point Homes in December 31, 1992. Both companies were builders of mid-scale and luxury homes in their respective markets. In 2004, because of tax considerations and the need to segment

the businesses between mid-scale and luxury homes, Beachwood decided to spin-off Country Point, its luxury home subsidiary, to its common shareholders. Beachwood retained Bernheim Securities to value the spin-off of Country Point as of December 31, 2004.

When the books closed on 2004, Beachwood had \$140 million in debt outstanding due in 2012 at a coupon rate of 8%, a spread of 2% above the current risk free rate. Beachwood also had 5 million common shares outstanding. It pays no dividends, has no preferred shareholders, and faces a tax rate of 30%. When valuing common stock, Bernheim's valuation models utilize a market risk premium of 11%.

The common equity allocated to Country Point for the spin-off was \$55.6 million as of December 31, 2004. There was no long-term debt allocated from Beachwood.

The Managing Director in charge of Bernheim's construction group, Denzel Johnson, is prepping for the valuation presentation for Beachwood's board with Cara Nguyen, one of the firm's associates. Nguyen tells Johnson that Bernheim estimated Country Point's net income at \$10 million in 2004, growing \$5 million per year through 2008. Based on Nguyen's calculations, Country Point will be worth \$223.7 million at the end of 2008. Nguyen decided to use a cost of equity for Country Point in the valuation equal to its return on equity at the end of 2004 (rounded to the nearest percentage point).

Nguyen also gives Johnson the table she obtained from Beachwood projecting depreciation (the only non-cash charge) and capital expenditures:

\$(in millions)	2004	2005	2006	2007	2008
Depreciation	5	6	5	6	5
Capital Expenditures	7	8	9	10	12

Looking at the numbers, Johnson tells Nguyen, "Country Point's free cash flow (FCF) will be \$25 million in 2006."

Nguyen adds, "That's FCF to the Firm (FCFF). FCF to Equity (FCFE) will be lower."

Question #73 - 76 of 137

Question ID: 1473042

Regarding the statements by Johnson and Nguyen about FCF in 2006:

- A) only Nguyen is incorrect.
- B) only Johnson is incorrect.
- C) both are incorrect.



Explanation

To estimate FCF, we can construct the following table using the table given and the information about growth in net income:

\$(in millions)	2004	2005	2006	2007	2008
Net Income	10	15	20	25	30
Plus: Depreciation	5	6	5	6	5
Less: Capital Expenditures	7	8	9	10	12
<i>Free Cash Flow</i>	8	13	16	21	23

The estimated free cash flow for 2006 is \$16 million. Johnson's statement is incorrect. Since none of Beachwood's debt is allocated to Country Point, all the financing is in the form of equity, so FCFF and FCFE are equal. Nguyen's statement is also incorrect.

Question #74 - 76 of 137

Question ID: 1473043

If FCInv equals Fixed Capital Investment and WCInv equals Working Capital Investment, which statement about FCF and its components is *least* accurate?

- A) $FCFE = (EBIT \times (1 - \text{tax rate})) + \text{Depreciation} - FCInv - WCInv$. 
- B) $FCFF = (EBITDA \times (1 - \text{tax rate})) + (\text{Depreciation} \times \text{tax rate}) - FCInv - WCInv$. 
- C) WCInv is the change in the working capital accounts, excluding cash and short-term borrowings. 

Explanation

The correct version of this equation is:

$$FCFF = (EBIT \times (1 - \text{tax rate})) + \text{Depreciation} - FCInv - WCInv$$

(Module 21.5, LOS 21.k)

Question #75 - 76 of 137

Question ID: 1473044

Given Nguyen's estimate of Country Point's terminal value in 2008, what is the growth assumption she must have used for free cash flow after 2008?

- A) 3%. 
- B) 7%. 
- C) 9%. 

Explanation

We know the terminal value in 2008 is \$223.7 million. We can calculate the free cash flow in 2008 to be \$23 million (= \$30 million net income + \$5 million depreciation – \$12 million capital expenditures). (See the table in question 1). Thus, we can solve for the estimated growth rate:

$$\text{Terminal value} = [CF@2008 \times (\text{growth rate} + 1)] / (\text{discount rate} - \text{growth rate})$$

$$223.7 \text{ million} = (\$23 \text{ million} \times (\text{growth rate} + 1)) / (0.18 - \text{growth rate})$$

$$223.7 \text{ million} \times (0.18 - \text{growth rate}) = 23 \text{ million} \times (\text{growth rate} + 1)$$

$$40.266 - (223.7 \times \text{growth rate}) = 23 \text{ million} + (23 \times \text{growth rate})$$

$$17.266 = 246.7 \times (\text{growth rate})$$

$$\text{growth rate} = 0.07$$

Nguyen's growth rate assumption is 7% per year

(Module 21.5, LOS 21.k)

Question #76 - 76 of 137

Question ID: 1473045

The value of beta for Country Point is:

A) 1.27.



B) 1.00.



C) 1.09.



Explanation

The risk free rate is $(8\% - 2\%) = 6\%$. We are told that the market risk premium is 11%, and we calculated the cost of equity (required return) to be $(10 \text{ million} / 55.6 \text{ million} =) 18\%$. Since we know the risk-free rate, the market risk premium, and the discount rate, we can use the capital asset pricing model to solve for beta:

$$\text{Required rate of return} = 0.18 = 0.06 + (b \times 0.11)$$

$$0.18 - 0.06 = b \times 0.11$$

$$0.12 = b \times 0.11$$

$$b = 1.09$$

(Module 21.5, LOS 21.k)

Question #77 of 137

Question ID: 1472967

The following information pertains to the Harrisburg Tire Company (HTC) in 2000.

- Earnings (net income) = \$600M.
- Dividends = \$120M.
- Interest expense = \$400M.
- Tax rate = 40%.
- Depreciation = \$500M.
- Capital spending = \$800M.
- Total assets = \$10B (book value and market value).
- Debt = \$4B (book value and market value).
- Equity = \$6B (book value and market value).

The firm's working capital needs are negligible, and they plan to continue to operate at their current capital structure.

The free cash flow to the firm is:

A) \$300M.



B) \$420M.



C) \$540M.



Explanation

The free cash flow to the firm is:

$$\text{FCFF} = \text{Net income} + (\text{Interest expense})(1 - T) - \text{Capital expenditures} + \text{Depreciation}$$

$$600\text{M} + 400\text{M}(1 - 0.40) - 800\text{M} + 500\text{M} = 540\text{M}$$

(Module 21.4, LOS 21.d)

Question #78 of 137

Question ID: 1472989

Currently, a firm has no outstanding debt. If the firm would add a small amount of leverage to its balance sheet, what should be the impact on the firm's value? There would be:

- A) no change in firm value. 
- B) an increase in value due to interest tax shields. 
- C) a decrease in value due to higher interest expense. 

Explanation

The amount of financial leverage used by a firm will affect its value. For small amounts of leverage, the additional bankruptcy risk will be low, and will be more than offset by the additional value of interest tax shields.

(Module 21.5, LOS 21.f)

Question #79 of 137

Question ID: 1472992

The repurchase of 20% of a firm's outstanding common shares will cause free cash flow to the firm (FCFF) to:

- A) remain the same. 
- B) decrease. 
- C) increase. 

Explanation

Share repurchases are a *use* of free cash flows, not a *source*. FCFF is cash flow that is available to all capital suppliers. Notice the conspicuous absence of repurchases in the following: $FCFF = CFO + Int(1 - \text{tax rate}) - FCInv$.

(Module 21.5, LOS 21.f)

Question #80 of 137

Question ID: 1586162

The following information is derived from the financial records of Brown Company for the year ended December 31, 2004:

Sales	\$3,400,000
Cost of Goods Sold (COGS)	(2,100,000)
Depreciation	(300,000)
Interest Paid	(200,000)
Gain on Sale of Old Equipment	400,000
Income Taxes Paid	<u>(300,000)</u>
Net Income	\$900,000

- Brown issued bonds on June 30, 2004 and received proceeds of \$4,000,000.
- Old equipment with a book value of \$2,000,000 was sold on August 15, 2004 for \$2,400,000 cash.
- Brown purchased land for a new factory on September 30, 2004 for \$3,000,000, issuing a \$2,000,000 note and paying the balance in cash.

Cash flow from operations less capital expenditures is:

A) \$200,000.



B) \$6,200,000.



C) \$2,200,000.



Explanation

Brown's cash flow from operations (CFO) was \$800,000 = (\$900,000 Net Income + \$300,000 depreciation - \$400,000 gain).

Capital expenditure cash flows were -\$3,000,000 for the factory and \$2,400,000 cash received from sale of the old equipment for a net outflow of cash of \$600,000.

$$\$200,000 = (\$800,000 - \$600,000).$$

(Module 21.1, LOS 21.c)

Question #81 of 137

Question ID: 1472996

In what ways are dividends different from free cashflow to equity (FCFE)?

A) There is no difference. Dividends must equal FCFE.



B) Dividends are often viewed as "sticky." Managers are reluctant to radically change the dividend payout policy while FCFE often has immense variability.



C) Companies often use FCFE as a signal of positive future growth prospects while dividends are not used for signaling.



Explanation

Dividends and the FCFE are often different and dividends are used as a signal to the market not FCFE. Dividends viewed as sticky is the true statement.

(Module 21.5, LOS 21.f)

Question #82 of 137

Question ID: 1472943

If a firm is valued using FCFF, the relevant discount rate is the:

- A) after-tax weighted average cost of capital.
- B) before-tax weighted average cost of capital.
- C) before-tax cost of equity.

**Explanation**

Since the FCFF is the cash available to all the investors, the after-tax weighted average cost of capital should be used as the discount rate in FCFF models.

(Module 21.1, LOS 21.a)

The following information was collected from the financial statements of Bankers Industrial Corp (BIC) for the year ended December 31, 2013.

- Earnings before interest and taxes (EBIT) = \$6.00 million.
- Capital expenditures = \$1.25 million.
- Depreciation expense = \$0.63 million.
- Working capital additions = \$0.59 million.
- Cost of debt = 10.50%.
- Cost of equity = 16.00%.
- Stable growth rate for FCFF = 7.00%.
- Stable growth rate for FCFE = 10.00%.
- Market value of debt = \$20.00 million.
- Book value of debt = \$22.50 million.
- Outstanding shares = 500,000.
- Interest expense = \$2.00 million.
- New Debt borrowing = \$3.30 million.
- Debt repayment = \$2.85 million.
- Growth rates for two-stage growth model for FCFE:
 - 25.0% for Years 1-3.
 - 6.0% for Years 4 and thereafter.

BIC is currently operating at their target debt ratio of 40.00%. The firm's tax rate is 40.00%.

Question #83 - 86 of 137

Question ID: 1473028

The free cash flow to the firm (FCFF) for the current year is *closest* to:

- A) \$2.39 million.
- B) \$2.31 million.
- C) \$3.57 million.

**Explanation**

The FCFF for the current year is $[\$6.00\text{m} \times (1 - 0.40)] + \$0.63\text{m} - \$1.25\text{m} - \$0.59\text{m} = \$2.39\text{m}$.

(Module 21.5, LOS 21.k)

Question #84 - 86 of 137

Question ID: 1473029

The estimated value of the firm is *closest* to:

- A) \$50 million.
- B) \$47 million.
- C) \$38 million.

**Explanation**

The value of BIC using a stable-growth FCFF model is \$49.95 million, calculated as:

$$\text{FCFF} = [\$6.00\text{m} \times (1 - 0.40)] + \$0.63\text{m} - \$1.25\text{m} - \$0.59\text{m} = \$2.39\text{m}$$

$$\text{WACC} = (0.60 \times 0.16) + [0.40 \times 0.105 \times (1 \times 0.40)] = 12.12\%.$$

$$\text{Estimated value} = (\$2.39\text{m} \times 1.07) / (0.1212 - 0.07) = \$49.95 \text{ million}.$$

(Module 21.5, LOS 21.k)

Question #85 - 86 of 137

Question ID: 1473030

If the estimated value of the free cash to the firm (FCFF) for year 0 is \$2.4 million, the value per share of BIC stock, based on the stable growth model, is *closest* to:

- A) \$39.
- B) \$55.
- C) \$61.

**Explanation**

$$\text{FCFE} = \text{FCFF} - \text{Interest expense} \times (1 - \text{tax rate}) + \text{Net borrowing} = \$2.40 \text{ million} - [\$2.00 \text{ million} \times (1 - 0.40)] + \$3.30 \text{ million} - \$2.85 \text{ million} = \$1.65 \text{ million}.$$

$$\text{The value of equity is: } [\$1.65 \text{ million} \times (1 + 0.10)] / (0.16 - 0.10) = \$30.25 \text{ million}.$$

$$\text{On a per share basis: } \$30.25 \text{ million} / 500,000 = \$60.50$$

(Module 21.5, LOS 21.k)

Question #86 - 86 of 137

Question ID: 1473031

The current market price of BIC is \$62.50 per share, and the current year's FCFE is \$1.75 million. Using a two-stage growth model to find the estimated the firm's value, the current market price BIC is most accurately described as:

- A) overvalued.
- B) undervalued.
- C) fairly valued.

**Explanation**

FCFE = FCFF – Interest expense × (1 – T) + New borrowing.

Year	0	1	2	3	4
Growth rate		25.0%	25.0%	25.0%	6.0%
FCFE in mil\$	\$1.750	\$2.188	\$2.734	\$3.418	\$3.623

The terminal value is $\$3,623 / (0.16 - 0.06) = \$36,230$ million. The calculator inputs: CF0 = 0, CF1 = \$2,188, CF2 = \$2,734, CF3 = \$3,418 + \$36,230 = \$39,648, I = 16, NPV = \$29.319 million.

Per share price is $\$29,319,000 / 500,000 = \58.64 . The stock appears to be overvalued at the current market price of \$62.50 per share, as our estimated value of \$58.64 suggests that the market price is too high.

(Module 21.5, LOS 21.k)

Question #87 of 137

Question ID: 1473035

A firm's free cash flow to equity (FCFE) in the most recent year is \$50M and is expected to grow at 5% per year forever. If its shareholders require a return of 12%, the value of the firm's equity using the single-stage FCFE model is:

- A) \$714M.
- B) \$417M.
- C) \$750M.



Explanation

The value of the firm's equity is: $\$50M \times 1.05 / (0.12 - 0.05) = \$750M$

(Module 21.5, LOS 21.k)

Question #88 of 137

Question ID: 1473022

BOX, Inc., earned \$4.55 per share last year. The firm had capital expenditures of \$1.75 per share and depreciation expense of \$1.05. BOX, Inc., has a target debt ratio of 0.25.

	High-Growth Period	Transitional Period	Stable-Growth Period
Duration	2 Years	5 Years	
Earnings growth rate	45%	Will decline 8% per year to 5% in the stable-growth period	5%
Growth in capital expenditures	30%	Increases by 8% per year	Same \$ amount as Depreciation
Growth in depreciation	30%	Increases by 13% per year	Same \$ amount as Capital Expenditures
Change in working capital	Given Below	Given Below	\$2.25 per share in Year 8
Shareholder required return	25%	15%	10%

	Yr 0	Yr 1	Yr 2	Yr 3	Yr 4	Yr 5	Yr 6	Yr 7
Earnings per share (EPS)	4.55	6.60	9.57	13.11	16.91	20.46	23.12	24.27
Capital expenditures	1.75	2.28	2.96	3.19	3.45	3.73	4.02	4.35
Depreciation	1.05	1.37	1.77	2.01	2.27	2.56	2.89	3.27
Change in working capital (WC)	0.90	1.10	1.40	1.60	1.80	2.00	2.20	2.10
Free cash flow to equity (FCFE)			7.63	11.01	14.67	18.08	20.62	21.89

What is the present value of BOX, Inc.?

A) \$223.65.



B) \$212.91.



C) \$195.71.



Explanation

Year 1 FCFE = Earnings per share – (Capital Expenditures – Depreciation)(1 – Debt Ratio) – (Change in working capital)(1 – Debt Ratio) = 6.60 – (2.28 – 1.37)(1 – 0.25) – (1.1)(1 – 0.25) = 5.09.

Year 8 FCFE = Earnings per share – (Capital Expenditures – Depreciation)(1 – Debt Ratio) – (Change in working capital)(1 – Debt Ratio) = 24.27 × 1.05 – 0 – (2.25)(1 – 0.25) = 23.79.

The Terminal Value (as of Year 7) = 23.79 / (0.10 – 0.05) = 475.80.

The value of BOX, Inc., stock would be equal to: $5.09 / 1.25 + 7.63 / 1.25^2 + 11.01 / [(1.25)^2(1.15)^1] + 14.67 / [(1.25)^2(1.15)^2] + 18.08 / [(1.25)^2(1.15)^3] + 20.62 / [(1.25)^2(1.15)^4] + 21.89 / [(1.25)^2(1.15)^5] + 475.80 / [(1.25)^2(1.15)^5] =$

$$4.07 + 4.88 + 6.13 + 7.10 + 7.61 + 7.55 + 6.97 + 151.40 = 195.71$$

(Module 21.5, LOS 21.k)

Question #89 of 137

Question ID: 1472987

An increase in financial leverage will cause free cash flow to equity (FCFE) to:

- A) increase in the year the borrowing occurred. 
- B) decrease or increase, depending on its circumstances. 
- C) decrease in the year the borrowing occurred. 

Explanation

An increase in financial leverage will increase net borrowing and, hence, increase FCFE in the year the borrowing occurred because: $FCFE = FCFF - [\text{interest expense}] (1 - \text{tax rate}) + \text{net borrowing}$.

(Module 21.5, LOS 21.f)

Question #90 of 137

Question ID: 1472988

Optimal capital structure is the mix of debt and equity that will maximize the value of the firm and minimize:

- A) the firm's cost of capital. 
- B) the amount of taxable profit reported. 
- C) agency costs of equity. 

Explanation

The optimal capital structure is the mix of debt and equity that will maximize the value of the firm and minimize weighted average cost of capital (i.e. the firm's cost of capital or "WACC").

(Module 21.5, LOS 21.f)

Question #91 of 137

Question ID: 1472945

Free cash flow (FCF) approaches are the *best* source of value when:

- A) FCFs track profitability closely over the analyst's forecast horizon. 
- B) a firm has preferred stock. 
- C) a firm is paying a dividend that is higher than the industry average. 

Explanation

FCF approaches are best when those flows are a good indication of a firm's profitability over the analyst's forecast horizon.

(Module 21.1, LOS 21.a)

Question #92 of 137

Question ID: 1472971

A firm currently has sales per share of \$10.00, and expects sales to grow by 25% next year. The net profit margin is expected to be 15%. Fixed capital investment net of depreciation is projected to be 65% of the sales increase, and working capital requirements are 15% of the projected sales increase. Debt will finance 45% of the investments in net capital and working capital. The company has an 11% required rate of return on equity. What is the firm's expected free cash flow to equity (FCFE) per share next year under these assumptions?

A) \$0.77.



B) \$1.88.



C) \$0.38.



Explanation

$FCFE = \text{net profit} - \text{NetFCInv} - \text{WCInv} + \text{DebtFin} = \$1.88 - \$1.63 - 0.38 + 0.90 = 0.77$

(Module 21.4, LOS 21.d)

Question #93 of 137

Question ID: 1473081

Sudbury Industries expects FCFF in the coming year of 400 million Canadian dollars (\$), and expects FCFF to grow forever at a rate of 3 percent. The company maintains an all-equity capital structure, and Sudbury's required rate of return on equity is 8 percent.

Sudbury Industries has 100 million outstanding common shares. Sudbury's common shares are currently trading in the market for \$80 per share.

Using the Constant-Growth FCFF Valuation Model, Sudbury's stock is:

A) overvalued.



B) undervalued.



C) fairly valued.



Explanation

Based on a free cash flow valuation model, Sudbury Industries shares appear to be fairly valued.

Since Sudbury is an all-equity firm, WACC is the same as the required return on equity of 8%.

The firm value of Sudbury Industries is the present value of FCFF discounted by using WACC. Since FCFF should grow at a constant 3 percent rate, the result is:

$$\text{Firm value} = \text{FCFF}_1 / \text{WACC} - g = 400 \text{ million} / 0.08 - 0.03 = 400 \text{ million} / 0.05 = \$8,000 \text{ million}$$

Since the firm has no debt, equity value is equal to the value of the firm. Dividing the \$8,000 million equity value by the number of outstanding shares gives the estimated value per share:

$$V_0 = \$8,000 \text{ million} / 100 \text{ million shares} = \$80.00 \text{ per share}$$

(Module 21.5, LOS 21.m)

Question #94 of 137

Question ID: 1472958

A control perspective is most consistent with which of the following valuation approaches?

A) Price to enterprise value.



B) Dividends.



C) Free cash flow (FCF).



Explanation

Dividend policy can be changed by the buyer of a firm. Thus, the FCF perspective looks to the source of dividends in a position of control rather than directly at dividends. The price to enterprise value approach does not focus on cash flows.

(Module 21.1, LOS 21.b)

Question #95 of 137

Question ID: 1472944

Valuation with free cash flow to equity and free cash flow to the firm:

A) both use the cost of equity.



B) use different discount rates.



C) both use the after-tax cost of debt.



Explanation

Free cash flow to the firm uses the weighted average cost of capital and free cash flow to equity uses the cost of equity. The key is to use a discount rate that reflects the opportunity cost of the indicated investor group.

(Module 21.1, LOS 21.a)

Question #96 of 137

Question ID: 1473023

SOX, Inc., expects high growth in the next 4 years before slowing to a stable future growth of 3%. The firm is assumed to pay no dividends in the near future and has the following forecasted free cash flow to equity (FCFE) information on a per share basis in the high-growth period:

	Year 1	Year 2	Year 3	Year 4
FCFE	\$3.05	\$4.10	\$5.24	\$6.71

High-growth period assumptions:

- SOX, Inc.'s, target debt ratio is 40% and a beta of 1.3.
- The long-term Treasury Bond Rate is 4.0%, and the expected equity risk premium is 6%.

Stable-growth period assumptions:

- SOX, Inc.'s, target debt ratio is 40% and a beta of 1.0.
- The long-term Treasury Bond Rate is 4.0% and the expected equity risk premium is 6%.
- Capital expenditures are assumed to equal depreciation.
- In year 5, earnings are \$8.10 per share while the change in working capital is \$2.00 per share.
- Earnings and working capital are expected to grow by 3% a year in the future.

What is the present value on a per share basis for SOX, Inc.?

A) \$70.49.



B) \$77.15.



C) \$64.24.



Explanation

The required rate of return in the high-growth period is $(r) = 0.04 + 1.3(0.06) = 0.118$.

The required rate of return in the stable-growth period is $(r) = 0.04 + 1.0(0.06) = 0.10$.

The Present Value (PV) of the FCFE in the high-growth period is $(3.05 / 1.118) + (4.10 / 1.118^2) + (5.24 / 1.118^3) + (6.71 / 1.118^4) = 14.06$.

The Terminal Price = Expected $FCFE_{n+1} / (r - g_n)$ with $FCFE_{n+1} = FCFE$ in year 5 = Earnings per share – (Capital Expenditures – Depreciation)(1 – Debt Ratio) – (Change in working capital)(1 – Debt Ratio) = $8.10 - 0(1 - 0.4) - 2.00(1 - 0.4) = 6.90$.

The Terminal Price = $6.90 / (0.10 - 0.03) = 98.57$.

The PV of the Terminal Price = $(98.57 / 1.118^4) = 63.09$.

The value of a share today is the PV of the FCFE in the high-growth period plus the PV of the Terminal Price = $14.06 + 63.09 = 77.15$.

(Module 21.5, LOS 21.k)

Question #97 of 137

Question ID: 1472982

The difference between the value estimate produced by the dividend discount model (DDM) and the one produced by the free cash flow to equity (FCFE) model can be accounted for by which of the following?

- A) Different estimates of model risk. 
- B) Different sales forecast. 
- C) The value in controlling the firm's dividend policy. 

Explanation

The difference between the value estimate produced by the DDM and the one produced by the FCFE model can be interpreted as the value of controlling the firm's dividend policy.

(Module 21.5, LOS 21.g)

Question #98 of 137

Question ID: 1473001

Which of the following types of companies is the two-stage free cash flow to equity (FCFE) model best suited for? Companies:

- A) growing at a rate similar to or less than the nominal growth rate of the economy. 
- B) with patents or firms in an industry with significant barriers to entry. 
- C) in high growth industries that will face increasing competitive pressures over time, leading to a gradual decline in growth to a stable level. 

Explanation

The two-stage model is best suited to analyzing firms in a high growth phase that will maintain that growth for a specific period, such as firms with patents or firms in an industry with significant barriers to entry. Companies growing at a rate similar to or less than the nominal growth rate of the economy are best suited for the single-stage FCFE Model. Companies in high growth industries correspond to the three-Stage FCFE Model.

(Module 21.5, LOS 21.j)

Harrisburg Tire Company (HTC) forecasts the following for 2013:

- Earnings (net income) = \$600M.
- Dividends = \$120M.
- Interest expense = \$400M.
- Tax rate = 40.0%.
- Depreciation = \$500M.
- Capital spending = \$800M.
- Total assets = \$10B (book value and market value).
- Debt = \$4B (book value and market value).
- Equity = \$6B (book value and market value).
- Target debt to asset ratio = 0.40.
- Shares outstanding = 2.0 billion

The firm's working capital needs are negligible, and HTC plans to continue to operate with the current capital structure. The tire industry demand is highly dependent on demand for new automobiles. Individual companies in the industry don't have much influence on the design of automobiles and have very little ability to affect their business environment. The demand for new automobiles is highly cyclical but demand forecast errors tend to be low.

Question #99 - 102 of 137

Question ID: 1473037

The firm's earnings growth rate is *most accurately* estimated as:

- A) 6.4%.
- B) 8.0%.
- C) 4.8%.



Explanation

The firm's estimated earnings growth rate is the product of its retention ratio and ROE:

$$g = RR \times (ROE) = [(600 - 120) / 600] \times (600 / 6000) = 0.08$$

(Module 21.5, LOS 21.k)

Question #100 - 102 of 137

Question ID: 1473038

The 2013 forecasted free cash flow to equity is:

- A) \$300M.
- B) \$340M.
- C) \$420M.



Explanation

Since working capital needs are negligible, the free cash flow to equity is:

$$FCFE = \text{Net income} - [1 - DR] \times [FCInv - \text{Depreciation}] - [(1 - DR) \times WCInv]$$

$$FCFE = 600M - [1 - 0.4] \times (800M - 500M) = 420M$$

where:

$$DR = \text{target debt to asset ratio}$$

(Module 21.5, LOS 21.k)

Question #101 - 102 of 137

Question ID: 1473039

If the total market value of equity is \$6.0 billion and the growth rate is 8.0%, the cost of equity based on the stable growth FCFE model is *closest* to:

- A) 7.0%.
- B) 15.0%.
- C) 14.0%.



Explanation

Value of equity = $FCFE_1 / (\text{Cost of equity} - \text{growth rate})$; so $\$6,000 = [\$420 / (\text{Cost of equity} - 0.08)]$

$(\text{Cost of equity} - 0.08) \times \$6,000 = \$420$

$\text{Cost of equity} - 0.08 = 0.07$

$\text{Cost of equity} = 0.15 = 15.0\%$

(Module 21.5, LOS 21.k)

Question #102 - 102 of 137

Question ID: 1473040

The beta for HTC is 1.056, the risk-free rate is 5.0% and the market risk premium is 10.0%.

The weighted average cost of capital for HTC is *closest* to:

A) 13.34%.



B) 15.56%.



C) 11.74%.



Explanation

$\text{Cost of equity} = r_f + (r_m - r_f) = 0.05 + 1.056(0.10) = 0.05 + 0.1056 = 0.1556$

The best approximation for cost of debt is the interest expense divided by the market value of the debt.

$\text{Cost of debt} = \text{Interest expense} / \text{market value of debt} = \$400 \text{ million} / \$4.0 \text{ billion} = 0.10$

$WACC = w_d \times r_d \times (1 - t) + w_e \times r_e = 0.40 \times 0.10 \times (1 - 0.40) + 0.60 \times 0.1556 = 0.1174$

(Module 21.5, LOS 21.k)

Question #103 of 137

Question ID: 1472941

Free cash flow to the firm valuation uses which discount rate?

A) Cost of equity.



B) After-tax cost of debt.



C) Weighted average cost of capital.



Explanation

Free cash flow to the firm valuation uses the opportunity cost relevant to the overall firm, which is the weighted average cost of capital.

(Module 21.1, LOS 21.a)

Question #104 of 137

Question ID: 1473000

Which of the following free cash flow to the firm (FCFF) models is *most* suited to analyze firms that are growing at a faster rate than the overall economy?

A) No growth FCFF model.



B) Two-stage FCFF model.



C) High growth FCFF model.



Explanation

The two-stage FCFF model is most suited for analyzing firms growing at a rate faster than the overall economy. The two-stage model assumes a high rate of growth for an initial period, followed by an immediate jump to a constant, stable growth rate.

(Module 21.5, LOS 21.j)

Jon Binkster, CFA, is researching a U.S. based company Basicomb Inc., who have just released their financial statements for the year ended December 20x5. These financial statements are included in Exhibits 1–3 below.

Exhibit 1

Basicomb Inc. Annual Income Statement

For the Year Ended December 31, 20x5 (in \$ millions)

Sales	721.9
Operating expenses	(417.0)
Operating profit	304.9
Gain on sale of fixed assets	9.6
Depreciation	(170.8)
Earnings before interest and tax	143.7
Interest expense	(40.3)
Pre-tax income	103.4
Income taxes	(31.0)
Net income	72.4

Exhibit 2

Basicomb Inc. Balance Sheet

As of December 31 (in \$ millions)

20x5	20x4
------	------

Current Asset	31.2	14.0
Cash and equivalents		
Accounts receivable	72.0	64.8
Inventories	<u>501.7</u>	<u>453.7</u>
Total current assets	604.9	532.5
<u>Non-Current Assets</u>		
Property, plant, and equipment	1138.7	982.7
Less: Accumulated depreciation	(370.0)	(216.0)
Net property, plant, and equipment	<u>768.7</u>	<u>766.7</u>
Total assets	<u>1373.6</u>	<u>1299.2</u>
<u>Current Liabilities</u>		
Accounts payable	60.1	62.5
Notes payable	<u>30.0</u>	<u>20.0</u>
Total current liabilities	90.1	82.5
<u>Non-Current Liabilities</u>		
Long term debt	<u>576.0</u>	<u>588.0</u>
Total liabilities	666.1	670.5
<u>Shareholders Equity</u>		
Common equity	384.0	360.0
Retained earnings	<u>323.5</u>	<u>268.7</u>
Total equity	<u>707.5</u>	<u>628.7</u>
Total liabilities and equity	<u>1373.6</u>	<u>1299.2</u>

Busicomb Inc. Cash Flow Statement**For the Year Ended December 31, 20x5 (in \$ millions)**Cash Flow from Operating Activities

Net income	72.4
Depreciation	170.8
Gain on sale of fixed assets	(9.6)

Change in Working Capital

(Increase) Decrease in accounts receivable	(7.2)
(Increase) Decrease in inventories	(48.0)
Increase (Decrease) in accounts payable	(2.4)
Net change in working capital	<u>(57.6)</u>
Net cash from operating activities	176.0

Cash Flow from Investing Activities

Purchase of property, plant, and equipment	(183.2)
Proceeds on disposal of plant and equipment	20.0
Net cash from investing activities	(163.2)
	12.8

Cash Flow from Financing Activities

Change in debt outstanding	(2.0)
Change in common stock	24.0
Payment of cash dividend	(17.6)
Net cash from financing activities	<u>4.4</u>

Net change in cash and cash equivalents	17.2
Cash at beginning of period	<u>14.0</u>
Cash at end of period	<u>31.2</u>

Busicomb Inc. announces a new strategic alliance that will require it to sell products on behalf of a very successful major European manufacturing and distribution operation. Binkster decides to value Busicomb using the dividend discount model (DDM) and the free cash flow-to-firm model (FCFF). Binkster undertakes a review of the financial performance of Busicomb Inc. using Exhibits 1 to 3 and forecasts related to the new sales contract, Binkster reaches the following conclusions:

- Earnings are expected to grow at 18%, next year before stabilizing to a long-term growth rate of 6% after six years (i.e., year 6). You can assume that the growth rate declines evenly each year between the high growth and the low growth period.
- The current payout ratio will be maintained, and the firm has a long-term target debt to capital ratio of 50%.
- Busicomb Inc. has an equity beta of 1.3.
- Government bond yield is 4.5% and the market equity risk premium is 6%.
- There are 12 million shares outstanding.
- The market value of the debt in the balance sheet of Busicomb Inc. is not materially different from the book value.
- The required rate of return of the debt holders is 7%.
- The effective tax rate of Busicomb is 30%.

Question #105 - 108 of 137

Question ID: 1473053

Using the three components of the DuPont system, and using opening balance sheet values, which of the following statements regarding 20x5 is correct?

- A)** Total asset turnover is 0.556, financial leverage is 2.07, and ROE is 11.5%. 
- B)** Total asset turnover is 1.798, financial leverage is 2.07, and ROE is 11.5%. 
- C)** Total asset turnover is 0.556, financial leverage is 0.483, and ROE is 11.5%. 

Explanation

ROE using the DuPont three component approach:

ROE = profit margin × asset turnover × financial leverage

$$\begin{aligned}
 &= \frac{\text{profit}}{\text{sales}} \times \frac{\text{sales}}{\text{assets}} \times \frac{\text{assets}}{\text{equity}} \\
 &= \frac{72.4}{721.9} \times \frac{721.9}{1299.2} \times \frac{1299.2}{628.7}
 \end{aligned}$$

$$= 0.100 \times 0.556 \times 2.066 = 0.1149 = 11.5\%$$

Note that it is typical to compute ROE based on opening equity for valuation contexts, whilst in FRA we typically use average values. The CFA Institute curriculum does note that some analyst compute ROE in a valuation context using average balance sheet values so care should be taken to follow the instructions in the question whether to use opening or average values.

Question #106 - 108 of 137

Question ID: 1473054

Assuming an ROE on 11.5%, which of the following is the best estimate of the sustainable growth rate for Basicomb Inc.?

A) 10.1%.



B) 8.7%.



C) 11.5%.

**Explanation**

Sustainable growth rate:

$g = \text{ROE} \times \text{earnings retention rate}$

$g = 11.54\% \times (1 - (17.6 / 72.4)) = 11.54\% \times 0.757 = 8.736\%$

(Module 20.3, LOS 20.p)

Question #107 - 108 of 137

Question ID: 1473055

Which of the following is the most accurate estimate of the value of a share of Basicomb Inc.'s common stock using the H model variant of the dividend discount model (DDM)? Work to the nearest \$ and assume the cost of equity is 12.3%.

A) \$33 per share.



B) \$32 per share.



C) \$42 per share.

**Explanation**

Cost of equity = $R_f + \beta(R_m - R_f)$

$= 4.5\% + 1.3(6\%) = 12.3\%$

Using the H model:

Dividend = $17.6\text{m} / 12\text{m} = \1.47

$$MV_0 = \frac{D_0(1+g_l)}{k-g_l} + \frac{D_0 \times H(g_h - g_l)}{k-g_l}$$
$$MV_0 = \frac{1.47(1+0.06)}{(0.123-0.06)} + \frac{1.47 \times \frac{6}{2}(0.18-0.06)}{(0.123-0.06)}$$

$MV = \$24.73 + \8.40

$MV = \$33.13$

(Module 20.3, LOS 20.n)

Question #108 - 108 of 137

Question ID: 1473056

It has also been estimated that the Free Cash Flow to the firm will be \$46m next year. It is expected to grow at 12% for the following two years before settling down to a long-term rate of growth of 6%. Which of the following is the *most* accurate estimate of the value of Basicomb Inc.'s business using a two stage FCFF model? Work to the nearest \$m.

A) \$2,402m.



B) \$2,204m.



C) \$1,968m.



Explanation

Using a two stage model:

$$\text{FCFF}_1 = \$46\text{m}$$

$$\text{WACC} = (12.3\% \times 0.5) + (7 \times (1 - 0.3) \times 0.5) = 8.6\%$$

$$\text{MV}_{\text{firm}} = \frac{46}{(1.086)^1} + \frac{46(1.12)}{(1.086)^2} + \frac{46(1.12)^2}{(1.086)^3} + \frac{46(1.12)^2(1.06)}{(0.086 - 0.06)(1.086)^3}$$

$$\text{MV}_{\text{firm}} = 42.36 + 43.68 + 45.05 + 1,836.69$$

$$\text{MV}_{\text{firm}} = 1967.78$$

(Module 21.5, LOS 21.k)

Michael Patrick, CFA, is a new analyst at EKS Inc., an investment management company based in New York. Patrick is attempting to value Johan Corp., a U.S. based company using free cash flows. Exhibits 1 through 2 present extracts from the financial statements, which are prepared according to U.S. GAAP.

Exhibit 1

Johan Corp.		
Summary Balance Sheets on 31 December (U.S. \$ millions)		
	2x12 Forecast	2x11 Actual
Cash	50	40
Accounts receivable	220	200
Inventory	<u>265</u>	<u>245</u>
Current assets	535	485
Gross PP&E	2,000	1,600
Accumulated depreciation	<u>(950)</u>	<u>(890)</u>
Total assets	1,585	1,195

Accounts payable	50	50
Short-term debt	<u>70</u>	<u>50</u>
Current liabilities	120	100
Long-term debt	620	400
Common stock	335	335
Retained earnings	<u>510</u>	<u>360</u>
Total liabilities and equity	1,585	1,195

Exhibit 2

Johan Corp.		
Summary Income Statement (U.S. \$ millions)		
	2x12 Forecast	2x11 Actual

EBIT	415	380
-------------	-----	-----

EBITDA	475	430
---------------	-----	-----

Interest expense	200	180
-------------------------	-----	-----

Tax rate = 30%

There will be no sales of long-term assets in 2x12.

Patrick who is unfamiliar with the free cash flow technique has been reviewing EKS's handbook for some guidance. The statements below are extracts from the handbook:

Statement 1: Share repurchases will reduce FCFE and will leave FCFF unchanged.

1:

Statement 2: Changes in leverage have a minor effect on FCFE and no effect on FCFF. For example, decreasing leverage through repayment of debt will decrease FCFE in the current year and increase forecasted FCFE in future years.

Statement 3: The loss on a sale of a long-term asset that has been included in operating expenses is a non-cash charge that should be adjusted for when calculating FCFF from EBIT.

Statement 4: The formula to calculate FCFF based on EBITDA is: $FCFF = EBITDA(1 - \text{tax rate}) - FC_{Inv} - WC_{Inv}$

4:

Patrick is also looking to value a potential takeover target, Fite Inc. Patrick has gathered the following data on Fite Inc. All figures are in millions of dollars.

	2x11	2x10	2x09	2x08
Net income	-\$12	\$110	\$50	\$95
FCFE	-\$5	-\$35	\$45	-\$20
FCFF	\$9	\$11	\$12	\$14
Dividends	\$10	\$10	\$9	\$9
Debt-to-equity	97%	94%	81%	85%

Patrick decides to use the most recent 2x11 FCFF of \$9 to estimate the value of the firm. He decides to use a three-stage model and makes the following assumptions:

High growth phase: FCFF will grow at 30% for the next three years. WACC = 18%

Transitional phase: For the following 3 years, growth will decline by 8% per year down to a stable growth rate of 6%. WACC = 15%

Stable-growth period: Growth will remain at 6% forever. WACC = 10%

Question #109 - 114 of 137

Question ID: 1473083

Calculate the forecasted free cash flow to the firm (FCFF) for 2x12, using the data in Exhibits 1 and 2.

A) -89.5.



B) -107.5.



C) -131.5.



Explanation

$$FCFF = EBIT \times (1 - \text{tax rate}) + \text{dep} - FC_{INV} - WC_{INV}$$

$$FCFF = 415 \times (1 - 0.30) + 60 - 400 - 40 = -89.5$$

Alternatively:

$$FCFF = [EBITDA \times (1 - \text{tax rate})] + (\text{dep} \times \text{tax rate}) - FC_{INV} - WC_{INV}$$

$$FCFF = [475 \times (1 - 0.30)] + (60 \times 0.30) - 400 - 40 = -89.5$$

Depreciation:

Accumulated depreciation 31 December 2011 = \$890

Accumulated depreciation 31 December 2012 = \$950

With no asset sales during the period the depreciation charge for 2012 = \$60

Fixed capital investment:

This is equal to capital expenditures (because there are no asset sales), which is equal to the change in gross PP&E:

$$FC_{INV} = 2000 - 1600 = 400$$

Working capital investment:

This is the change in the working capital accounts, excluding cash and short-term borrowings.

Working capital = receivables + inventory – payables

$$WC_{INV} = (220 + 265 - 50) - (200 + 245 - 50) = 40$$

(Module 21.4, LOS 21.d)

Question #110 - 114 of 137

Question ID: 1473084

Calculate the forecasted free cash flow to equity (FCFE) for 2x12.

A) 10.5.



B) -9.5.



C) -49.5.



Explanation

$$FCFE = FCFF - \text{Int} (1 - \text{tax rate}) + \text{net borrowing}$$

$$FCFE = -89.5 - 200(1 - 0.3) + 240 = 10.5$$

Net borrowing is the difference between the long-term and short-term debt accounts (the impact caused by amortization of premiums and discounts should be removed if debt is not issued at par):

$$= (620 + 70) - (400 + 50) = 240$$

Alternatively:

$$FCFE = NI + \text{dep} - FC_{INV} - WC_{INV} + \text{net borrowing}$$

$$= [(415 - 200) \times 0.7] + 60 - 400 - 40 + 240 = 10.5$$

(Module 21.4, LOS 21.d)

Question #111 - 114 of 137

Question ID: 1473085

Regarding the handbook's statements on free cash flow techniques:

	<u>Statement 1</u>	<u>Statement 2</u>	
A)	Correct	Correct	✗
B)	Incorrect	Correct	✓
C)	Correct	Incorrect	✗

Explanation

Dividends, share repurchases, and share issues have no effect on either FCFF or FCFE. FCFF and FCFE represent the total cash flow available before financing decisions. Share repurchases represent uses of those cash flows.

If debt is repaid FCFE, will decrease in the current year because of reduced (negative) net borrowings and will increase in future years as interest expense is reduced.

(Module 21.1, LOS 21.b)

Question #112 - 114 of 137

Question ID: 1473086

Regarding the handbook's statements on free cash flow techniques:

	<u>Statement 3</u>	<u>Statement 4</u>	
A)	Correct	Correct	✗
B)	Incorrect	Correct	✗
C)	Correct	Incorrect	✓

Explanation

$$FCFF = [EBIT \times (1 - \text{tax rate})] + \text{dep} - FC_{INV} - WC_{INV}$$

We assume that the only non-cash charge that appears above EBIT is depreciation. In general however the rule is to adjust for any non-cash charges that appear above EBIT.

$$FCFF = [EBITDA \times (1 - \text{tax rate})] + (\text{dep} \times \text{tax rate}) - FC_{INV} - WC_{INV}$$

(Module 21.4, LOS 21.d)

Question #113 - 114 of 137

Question ID: 1473087

The *most appropriate* model for valuing Fite Inc. is the:

- | | | |
|----|-----------------------------------|---|
| A) | free cash flow to equity model. | ✗ |
| B) | free cash flow to the firm model. | ✓ |
| C) | dividend discount H-model. | ✗ |

Explanation

A dividend discount model is inappropriate, as dividends are not related to the earnings stream. In addition, as this is a takeover situation a free cash flow approach is more suitable as the acquirer has control and discretion over the distribution of the total free cash flow. With dividend discount models a minority, interest is assumed (i.e., no control over dividend policy).

FCFF model is preferred to FCFE as FCFE is negative and volatile and leverage is high.

(Module 21.1, LOS 21.b)

Question #114 - 114 of 137

Question ID: 1473088

Assuming Patrick is correct to use free cash flow to the firm to value Fite Inc.; the value of the firm is *closest* to:

A) 379.



B) 412.



C) 22.



Explanation

High Growth Period			
	Year 1	Year 2	Year 3
Growth rate	30%	30%	30%
FCFF	11.7	15.21	19.773
PV(@18%)	9.915	10.924	12.034

Transitional Period	Year 4	Year 5	Year 6
Growth rate	22%	14%	6%
FCFF	24.123	27.500	29.150
PV(@18%/15%)	$* \frac{1}{1.15 * 1.18^3}$ = 12.767	$* \frac{1}{1.15^2 * 1.18^3}$ = 12.656	$* \frac{1}{1.15^3 * 1.18^3}$ = 11.666

Terminal value as of Year 6 using the FCFF projected for Year 7.

Terminal value = $29.150 (1.06) / (0.10 - 0.06) = 772.48$

PV of terminal value = $772.48 / (1.15^3 \times 1.18^3) = 309.135$

Value of the firm = $9.915 + 10.924 + 12.034 + 12.767 + 12.656 + 11.666 + 309.135 = 379$

(Module 21.5, LOS 21.k)

Michael Ballmer is an equity analyst with New Horizon Research. The firm has historically relied on dividend and residual income valuation models to value equity, but the firm's director of research, Doug Leads, has decided that the firm needs to incorporate free cash flow valuations into its practices. Therefore, Leads decides to send Ballmer to a seminar on free cash flow valuation.

Upon his return from the convention, Ballmer is excited to share his newfound knowledge with his co-workers. Ballmer is asked to give a debriefing to New Horizon's team of equity analysts, where he makes the following statements:

- Statement 1: Free cash flow to the firm is the amount of the firm's cash flow that is free for the firm to use in making investments after cash operating expenses have been covered.
- Statement 2: Free cash flow to equity, then, is the amount of the firm's cash flow that is free for equity holders after covering cash operating expenses, working capital and fixed capital investments, interest principal payments to bondholders, and required dividend payments.

After discussing the calculation of free cash flow to the firm and free cash flow to equity from historical information, Ballmer proceeds to explain the major approaches for forecasting free cash flow. He focuses his discussion on forecasting the components of free cash flow as this method is more flexible. During his presentation, several of the analysts notice that the formula for forecasting free cash flow to equity does not include net borrowing. They bring this to Ballmer's attention, and he states that he will look into the formula and send out an updated presentation after the meeting.

A week after the meeting, Jonathan Hodges approached Ballmer regarding two issues he had while applying free cash flow based valuations. The first issue that Hodges had was that he calculated the equity value of a firm using both free cash flow to equity based and dividend-based valuations and arrived at different values. The second issue that Hodges came across was the effect of a change in a firm's target leverage on FCFE. One of the firms that Hodges was analyzing may reduce leverage, and Hodges needs to know if this will affect his valuation.

Question #115 - 118 of 137

Question ID: 1472953

Regarding statements 1 and 2, are Ballmer's interpretations of free cash flow to the firm (FCFF) and free cash flow to equity (FCFE) CORRECT?

- A) No, only one interpretation is correct.
- B) Yes, both interpretations are correct.
- C) No, neither interpretation is correct.



Explanation

Free cash flow to the firm (FCFF) is the cash flows that are free to investors after cash operating expenses (including taxes but excluding interest expense), working capital investments, and fixed capital investments have been made. Free cash flow to equity (FCFE) is FCFF less interest payments to bondholders and net borrowing from bondholders.

(Module 21.1, LOS 21.a)

Question #116 - 118 of 137

Question ID: 1472954

Which of the following statements regarding forecasting FCFE using the components of free cash flow method and net borrowing is *most* accurate?

- A) Net income already accounts for interest expense; therefore, net borrowing is not needed. 
- B) Investment in fixed capital and net borrowing are assumed to offset each other. 
- C) The target debt-to-asset ratio accounts for the financing of new investment in fixed capital and working capital. 

Explanation

When forecasting FCFE, it is common to assume that a firm will maintain a target debt-to-asset ratio for new investments in fixed capital and working capital. Based on this assumption, the formula for forecasting FCFE is:

$$FCFE = NI - [(1 - DR) \times (FCInv - Dep)] - [(1 - DR) \times WCInv]$$

By multiplying the fixed capital and working capital investments by one minus the target debt-to-asset ratio, you are left with the investment amount less the amount financed by debt, which is the net borrowing amount. Therefore, this formula accounts for net borrowing through the target debt-to-asset ratio.

(Module 21.1, LOS 21.a)

Question #117 - 118 of 137

Question ID: 1472955

Should dividend-based and free cash flow from equity (FCFE) based valuations result in different equity values for a firm?

- A) Yes, dividend-based valuations would be higher for firms with large, consistent dividends. 
- B) No, both models should result in the same value. 
- C) Yes, the free cash flow from equity valuation would be higher if there were a premium associated with control of the firm. 

Explanation

The ownership perspectives of dividend-based and FCFE based valuations are different. Dividend-based valuations take the perspective of minority shareholders, while FCFE based valuations take the perspective of an acquirer who will assume a controlling position in the firm. If investors were willing to pay a premium for a controlling position in the firm, then the equity value computed under the FCFE approach would be higher.

(Module 21.1, LOS 21.a)

Question #118 - 118 of 137

Question ID: 1472956

Which of the following statements regarding the effect a decrease in leverage has on a firm's free cash flow from equity (FCFE) is *most* accurate?

- A) Current year FCFE decreases, but future FCFE will be increased. 
- B) FCFE is unaffected by changes in leverage. 
- C) Current year FCFE increases, but future FCFE will be reduced. 

Explanation

Changes in leverage do have a small effect on FCFE. A decrease in leverage will cause the current year FCFE to decrease through the repayment of debt. Future FCFE will be increased because interest expense will be lower.

(Module 21.1, LOS 21.a)

Question #119 of 137

Question ID: 1472947

When using the two-stage FCFE model, if increases in working capital appear too high the analyst should:

- A) switch to a three-stage model. 
- B) use changes that are based upon a working capital ratio that is closer to the industry average. 
- C) normalize them to be equal to zero. 

Explanation

The best solution is to use changes that are based upon a working capital ratio that approximates the industry average. The problem will not be eliminated by switching to a three-stage FCFE model.

(Module 21.1, LOS 21.a)

Question #120 of 137

Question ID: 1473074

Terminal value in multi-stage free cash flow valuation models is often calculated as the present value of:

- A) a two-stage valuation model's price. 
- B) free cash flow divided by the growth rate. 
- C) a constant growth model's price as of the beginning of the last stage. 

Explanation

Terminal values are usually calculated as the present value of the price produced by a constant-growth model as of the beginning of the last stage.

(Module 21.5, LOS 21.I)

Question #121 of 137

Question ID: 1472997

If the investment in fixed capital and working capital offset each other, free cash flow to the firm (FCFF) may be proxied by:

- A) earnings before interest and taxes (EBIT). 
- B) net income plus non-cash charges plus after-tax interest. 
- C) net income plus after-tax interest. 

Explanation

The answer is indicated by the definition of FCFF: $FCFF = NI + NCC + Int(1 - \text{tax rate}) - FCInv - WCInv$. The relationship between net income and FCFF is indicated by: $NI = EBIT(1 - \text{tax rate}) - Int(1 - \text{tax rate})$.

(Module 21.5, LOS 21.h)

Question #122 of 137

Question ID: 1472957

An analyst is performing an equity valuation for a minority equity position in a dividend paying multinational. The appropriate model for this analysis is *most likely*:

- A) FCFF approach. 
- B) FCFE approach. 
- C) The Dividend Discount approach. 

Explanation

The dividend discount model is most appropriate for valuing a minority equity position in a dividend-paying company. The free cash flow approach looks to the source of dividends from the perspective of an owner that has control rather than directly at dividends.

(Module 21.1, LOS 21.b)

Question #123 of 137

Question ID: 1473026

The value of stock under the two-stage FCFE model will be equal to:

- A) present value (PV) of FCFE during the extraordinary growth period plus the terminal value. 
- B) present value (PV) of FCFE during the extraordinary growth and transitional periods plus the PV of terminal value. 
- C) present value (PV) of FCFE during the extraordinary growth period plus the PV of terminal value. 

Explanation

The value of stock under the two-stage FCFE model will be equal to the *present value* of FCFE during the extraordinary growth period plus the *present value* of the terminal value at the end of this period.

(Module 21.5, LOS 21.k)

Question #124 of 137

Question ID: 1473072

In the two-stage FCFE model, the required rate of return for calculating terminal value should be:

- A) lower than the required rate of return used for the high-growth phase. 
- B) higher than the required rate of return used for the high-growth phase. 
- C) equal to the average required rate of return for the industry. 

Explanation

In most cases, the required rate of return used to calculate the terminal value should be lower than the required rate of return used for initial high-growth phase. During the stable period the firm is less risky and the required rate of return is therefore lower.

(Module 21.5, LOS 21.i)

Question #125 of 137

Question ID: 1473013

A three-stage free cash flow to the firm (FCFF) is typically appropriate when:

- A) growth is currently low and will move through a transitional stage to a final stage wherein growth exceeds the required rate of return. 
- B) growth is currently high and will move through a transitional stage to a steady-state growth rate. 
- C) the required rate of return is less than the growth rate in the last stage. 

Explanation

The three-stage model using either FCFE or FCFF typically assumes that growth is currently high and will move through a transitional stage to a steady-state growth rate. Multi-stage models assume that the required rate of return exceeds the growth rate in the last stage.

(Module 21.5, LOS 21.j)

Question #126 of 137

Question ID: 1473070

A firm has:

- Free cash flow to the firm = \$4.0 million.
- Weighted average cost of capital = 10%.
- Total debt = \$30.0 million.
- Long-term expected growth rate = 5%.
- Value of the firm = \$50.00 per share.

What will happen to the value of the firm if the weighted average cost of capital increases to 12%?

- A) The value will increase. 
- B) The value will decrease. 
- C) The value will remain the same. 

Explanation

Everything else being constant, an increase in the relevant required rate of return should decrease the value of the firm.

(Module 21.5, LOS 21.i)

Question #127 of 137

Question ID: 1473025

Industrial Light currently has:

- Free cash flow to equity = \$4.0 million.
- Cost of equity = 12%.
- Weighted average cost of capital = 10%.
- Total debt = \$30.0 million.
- Long-term expected growth rate = 5%.

What is the value of equity?

- A)** \$57,142,857.
- B)** \$27,142,857.
- C)** \$60,000,000.



Explanation

The value of equity is $[(\$4,000,000)(1.05) / (0.12 - 0.05)] = \$60,000,000$.

(Module 21.5, LOS 21.k)

Question #128 of 137

Question ID: 1472948

The difference between free cash flow to equity (FCFE) and free cash flow to the firm (FCFF) is:

- A)** after-tax interest and net borrowing.
- B)** earnings before interest and taxes (EBIT) less taxes.
- C)** before-tax interest and net borrowing.



Explanation

$FCFE = FCFF - [\text{interest expense}] (1 - \text{tax rate}) + \text{net borrowing}$.

(Module 21.1, LOS 21.a)

Question #129 of 137

Question ID: 1473069

A firm has:

- Free cash flow to equity = \$4.0 million.
- Cost of equity = 12%.
- Long-term expected growth rate = 5%.
- Value of equity per share = \$57.14 per share.

What will happen to the value of equity if the cost of equity decreases to 10%?

- A)** The value will increase.
- B)** The value will decrease.
- C)** There is insufficient information to tell.



Explanation

Everything else being constant, a decrease in the relevant required rate of return should increase the value of the equity per share.

(Module 21.5, LOS 21.i)

Question #130 of 137

Question ID: 1472966

A firm currently has the following per share values:

- Cash flow from operations (CFO) is \$49.50.
- Investment in fixed capital is \$40.00.
- Net borrowing is \$7.50.

What is the current per share free cash flow to equity (FCFE)?

A) \$16.50.



B) \$97.00.



C) \$17.00.



Explanation

$FCFE = CFO - FCInv + \text{net borrowing} = \$49.50 - \$40.00 + \$7.50 = \$17.00$

(Module 21.4, LOS 21.d)

Question #131 of 137

Question ID: 1472976

A common approach to forecasting free cash flows is to:

A) project net income and expected capital expenditures.



B) project earnings before interest and taxes (EBIT) and expected capital expenditures.



C) calculate historical free cash flow and apply an expected growth rate.



Explanation

Historical free cash flows are often used for forecasting.

(Module 21.5, LOS 21.e)

Question #132 of 137

Question ID: 1472975

The following table provides forecasts for next year on a per share basis for TOY Inc.:

Item	Forecast
Earnings	\$5.00
Capital Expenditures	\$2.40
Depreciation	\$1.80
Change in Working Capital	\$1.70

TOY Inc.'s target debt ratio is 30% and has a required rate of return of 12%. Earnings, capital expenditures, depreciation, and working capital are all expected to grow by 5% a year in the future. Assume that capital expenditures and working capital are financed at the target debt ratio.

What is the forecasted free cashflow to equity (FCFE) for TOY Inc.?

A) \$4.31.



B) \$2.70.



C) \$3.39.



Explanation

$FCFE = \text{Earnings per share} - (\text{Capital Expenditures} - \text{Depreciation})(1 - \text{Debt Ratio}) - \text{Change in working capital}(1 - \text{Debt Ratio})$
 $= 5.00 - (2.40 - 1.80)(1 - 0.3) - (1.7)(1 - 0.3) = 3.39.$

(Module 21.5, LOS 21.e)

Question #133 of 137

Question ID: 1472993

The repayment of a significant amount of outstanding debt will cause free cash flow to equity (FCFE) to:

A) remain the same.



B) increase.



C) decrease.



Explanation

Debt repayment will decrease net borrowing and, hence, decrease FCFE because: $FCFE = FCFF - [\text{interest expense}](1 - \text{tax rate}) + \text{net borrowing}.$

(Module 21.5, LOS 21.f)

Question #134 of 137

Question ID: 1473073

In five years, a firm is expected to be operating in a stage of its life cycle wherein its expected growth rate is 5%, indefinitely; its required rate of return on equity is 11%; its weighted average cost of capital is 9%; and the free cash flow to equity in year 6 will be \$5.25 per share. What is its projected terminal value at the end of year 5?

A) \$51.93.



B) \$87.50.



C) \$131.25.



Explanation

Terminal value = $FCFE / (k - g) = \$5.25 / (0.11 - 0.05) = \87.50

(Module 21.5, LOS 21.l)

Question #135 of 137

Question ID: 1473024

A firm's free cash flow to the firm (FCFF) in the most recent year is \$80M and is expected to grow at 3% per year forever. If the firm has \$100M in debt financing and its weighted average cost of capital is 10%. The value of the firm's equity using the single-stage FCFF model is:

A) \$1,177M.



B) \$1,077M.



C) \$1,043M.



Explanation

The value of the firm's equity is equal to the value of the firm minus the value of the debt. Firm value = $\$80M \times 1.03 / (0.10 - 0.03) = \$1,177M$, so equity value is $\$1,177M - \$100M = \$1,077M$.

(Module 21.5, LOS 21.k)

Question #136 of 137

Question ID: 1473010

A firm in stable growth phase should have:

A) a required rate of return close to the market rate of return and capital expenditures that are not too large relative to depreciation expense.



B) capital expenditures that are less than the depreciation expense.



C) a growth rate higher than that of the economy and a required rate of return that is greater than the market rate of return.



Explanation

A firm that is in a stable growth phase should have growth rate close to that of the economy, and the cost of equity should approximate the required rate of return on the market. In addition, the capital expenditures should not be disproportionately large relative to the depreciation expense.

(Module 21.5, LOS 21.j)

Question #137 of 137

Question ID: 1473034

Industrial Light currently has:

- Expected free cash flow to the firm in one year = \$4.0 million.
- Cost of equity = 12%.
- Weighted average cost of capital = 10%.
- Total debt = \$30.0 million.
- Long-term expected growth rate = 5%.

What is the value of equity?

A) \$50,000,000.



B) \$80,000,000.



C) \$44,440,000.



Explanation

The overall value of the firm is $\$4,000,000 / (0.10 - 0.05) = \$80,000,000$. Thus, the value of equity is $\$80,000,000 - \$30,000,000 = \$50,000,000$.

(Module 21.5, LOS 21.k)